

DeadEye handbook

pre-publication guide, research plan, pre-registration, peer testing, budget, limitations, references audit

8 October 2026

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Before you publish

This page is written for the person who will run the study and submit the paper. It says plainly what exists today, what is still missing, and the order in which to do the remaining work.

What is done

- The benchmark software is complete and tested: seven decision tasks with known optimal policies, four model backends (local Hugging Face weights, OpenAI-compatible servers, decision-model APIs, and a mock for tests), six conversion methods, a resumable runner that logs every decision, statistics (confidence intervals, paired tests, scale slopes, calibration), and a report generator that writes every table and figure the paper uses. `pytest` and `deadeye` smoke pass.
- The paper is drafted in full: abstract, introduction, related work, benchmark, setup, results structure, analysis plan, limitations, conclusion and appendix (every prompt verbatim), with a verified Harvard bibliography of 124 entries.
- The research plan, pre-registration, compute budget, peer-testing guide and known limitations are written.

What is not done

- **No real model has been evaluated.** The sandbox in which this was built had no GPU and no access to Hugging Face, so every experiment ran only on a mock model and a tiny randomly initialised model, which prove the pipeline works and nothing else. The Metal backend was added for your Mac but could not be exercised here; the first thing the pilot will tell you is whether it loads and runs, and the fallback is `dtype: float16` or `device: cpu` in the config.
- **Every number in the paper's results is a placeholder.** Placeholders are the blue `\result{...}` marks (127 of them) and the red `\todo{...}` notes (23). The draft PDF prints them in colour so none can be missed.
- **The pre-registration is not frozen** (`docs/preregistration.md` still has blanks for the pilot's standard deviations), and nothing has been tagged.
- **Laya may need a small adapter** before it can be scored if its HTTP schema differs from the System One protocol; Kev speaks the protocol already implemented, and `SemIf` is reproduced by the letter-scoring row.
- **Two small source checks** remain: the TechCrunch bylines in the bibliography, and the Qwen3.5 repository ids in `configs/decision_models.yaml`, which must be confirmed on the Hugging Face hub.
- **Everything the laptop cannot run is listed in `docs/future_work.md`** and in the paper's future-work section: the 14B to 72B rungs, the hosted decision models, long thinking budgets, and more.

What to do, in order

Each step says what you need, what to run, how long it takes, and what you should see. Everything runs on your Mac; no GPU rental is needed, and none of these commands uses Claude or any paid service.

Step 1. Install on the Mac (one hour)

```
git clone https://github.com/CaptiveON/DeadEye && cd DeadEye
python3 -m venv .venv && source .venv/bin/activate
pip install torch # the default macOS wheel includes the Metal (MPS) backend
pip install -e ".[hf,dev]" tabulate
pytest -q && deadeye smoke # about five minutes; all tests pass, a smoke report is written
huggingface-cli login # free account; needed for the gated Gemma 3 repositories
python -c "import torch; print(torch.backends.mps.is_available())" # must print True
```

Accept the Gemma licence on the Hugging Face page of `google/gemma-3-270m` once; the other families are open.

Step 2. Run the pilot (3 to 6 hours)

```
scripts/run_mac.sh pilot
deadeye report results/pilot --out report/pilot
```

Open `report/pilot/index.html`. Check: random scores 0 and oracle 1 on every task; the illegal-action rate under `prompt_generate` is below about 20% for the instruct models (if not, read the raw outputs in `steps.jsonl` and fix the wording in `src/deadeye/policies/prompts.py`); the probe's `train_acc` in `summary.json` is well above chance. Note the measured `latency_per_decision_s` per model size and re-estimate the sweep with `deadeye estimate configs/mac_main.yaml --seconds-per-decision <value>`. Prompts may be edited only now.

Step 3. Freeze the design (one hour) Fill in the per-task standard deviations in section 5 of `docs/preregistration.md` from `report/pilot/episodes.csv` (the standard deviation of the paired differences of norm between two methods on the same seeds), then:

```
git add -A && git commit -m "Freeze pre-registration" && git tag prereg-v1 && git push --tags
```

Step 4. Run the main sweep (two to three weeks of overnight runs)

```
scripts/run_mac.sh main
```

The ladders run in the order Qwen2.5, Qwen3, SmolLM2, Gemma 3, so a stopped run still leaves complete ladders. Rerun the same command after any interruption; finished cells are skipped. `logs/mac_main.log` has the progress line of every cell, and `results/mac_main/run_manifest.json` the full cell list.

Step 5. LoRA, free replies, controls (one to two weeks of overnight runs)

```
scripts/run_mac.sh lora
scripts/run_mac.sh free
scripts/run_mac.sh controls    # optional; Pythia and OLMo 2
```

Step 6. Ablations (about a week)

```
scripts/run_mac.sh ablations    # prompting, reasoning, adaptation, tasks, tasks_lora
brew install llama.cpp          # for the quantisation block; download the Q8_0 and Q4_K_M GGUF files of
                                # Qwen2.5-Instruct 1.5B, 3B, 7B, start one llama-server per file on the ports in
                                # configs/ablations/quantisation.yaml, then:
scripts/run_mac.sh quant
```

Step 7. Decision models (a day) Install Kev (github.com/jaredpalmer/kev) and start its server for the 0.8B and 4B checkpoints on ports 8009 and 8010; install Laya's package and SemIf as their repositories describe; confirm the two Qwen3.5 base repository ids on the hub and correct them in the config if needed. If Laya's HTTP schema differs from the System One protocol, add the small adapter noted in `src/deadeye/models/decision_api.py`. Then:

```
scripts/run_mac.sh decision
```

Step 8. Generate every table and figure (minutes)

```
deadeye report results/mac_main results/mac_lora results/mac_free_reply results/mac_controls \
results/ablations results/decision_models --out report/paper --paper-dir paper
```

`report/paper/cells.csv`, `slopes.csv`, `pairwise_methods.csv` and `system_comparison.csv` hold every number the paper quotes; `paper/tables/*.tex` and `paper/figures/*.pdf` are included by the paper automatically.

Step 9. Fill the results (two to three days of careful work) Work through `paper/sections/results.tex`, `analysis.tex`, `abstract.tex`, `introduction.tex` and `conclusion.tex`. Replace each `\result{...}` with the measured value and its confidence interval from the CSV files, delete each `\todo{...}` once handled, run one `deadeye compare` call per hypothesis family and quote the Holm-adjusted p-values, then write "supported" or "falsified" against each pre-registered criterion. A falsified hypothesis is a result.

```
grep -c '\\result{' paper/sections/*.tex    # must reach 0
grep -c '\\todo{'  paper/sections/*.tex    # must reach 0
python scripts/check_tex.py                 # braces, environments, citation keys
```

Step 10. Final checks, PDF, submission (a day) Add the TechCrunch bylines to `paper/refs.bib`; reopen each model card cited in the decision-model table and confirm sizes and licences; rerun `python scripts/check_tex.py`. Build the PDF with `cd paper && make (TeX)` or `python scripts/build_draft_pdf.py` (no TeX needed). Archive `results/` and `report/` with a DOI (Zenodo), tag v1.0, submit to arXiv and then the venue.

Honest expectations

The design was reviewed adversarially during construction and every oracle was checked, but no study survives contact with real models unchanged. Expect to find a prompt that a model family misreads, a checkpoint that fails to load, or a task whose variance is higher than planned. All of these are handled by the plan: prompts change only before the tag, failed cells are listed in the appendix, and the power section reports the smallest effect the design could detect instead of claiming a null.

Research plan

The question we want to answer is simple to state and surprisingly open: when you take an open-weight language model and turn it into something that makes decisions, what actually determines how good those decisions are? Everyone assumes bigger is better. But "turning a model into a decision-maker" can mean prompting it, scoring its likelihoods, training a small head on its hidden states, or fine-tuning it on demonstrations, and these use the weights in very different ways. Our bet is that the conversion method and the shape of the task matter at least as much as the parameter count, and that a small model handled well will often match a much larger one handled naively. The experiments below are designed to find out whether that bet holds, and where it fails.

Three things come out of the project. The first is the deadeye harness in this repository: a benchmark that anyone can install with pip, run on a laptop for a pilot or on a GPU box for the full sweep, and that produces a leaderboard from its own logs. The second is a paper (skeleton in `paper/`) in which every table and figure is generated by `deadeye report`, so nothing in it is typed by hand. The third is a pre-registration (`docs/preregistration.md`) that pins down the hypotheses and the analysis before the main sweep runs, so that we cannot quietly move the goalposts afterwards.

The whole study runs on one Apple M2 Max laptop with 32 GB of memory, through PyTorch's Metal backend at 16-bit precision. That decides the scope: every method, including probes and LoRA, up to about 8B parameters (LoRA to 4B), four ladders from 135M to 8B, and nothing that needs a data-centre GPU. What the laptop excludes (the 14B to 72B rungs, the hosted decision models, long thinking budgets) is listed in `docs/future_work.md` and in the paper's future-work section, with the hardware and the approximate cost each item would take.

What follows is the whole route: where to start, what each phase produces, how to test it, and what "finished" looks like. Every command in this document runs today.

1. Framing

1.1 What "converting a weighted model into a decision model" means here A decision model is a policy: it maps a state or observation to an action. A language model is not one; it is a distribution over text. Bridging the two is where most of the design choices hide, and it is usually done one way, without comment, in papers that then draw conclusions about "the model". We treat the conversion method as an experimental factor in its own right, with five levels:

Method (<code>methods[].name</code>)	What is used from the model	Training data	Where it runs
<code>prompt_free</code>	the unconverted assistant: it answers the situation in its own words and an action is extracted from the reply if one can be; the reference point for "before conversion" latency, tokens and usable-decision rate	none	any backend
<code>prompt_generate</code>	free-form generation, parsed into an action (zero-shot, few-shot, chain-of-thought variants)	none	any backend
<code>prompt_score</code>	exact log-likelihood of each legal action as a continuation (lm-eval-harness style)	none	hf; first-token approximation on <code>openai</code>
<code>probe</code>	hidden state of the prompt at one layer, with a logistic-regression head	oracle-labelled states from training seeds	hf
<code>lora_sft</code>	LoRA behaviour cloning on oracle (observation, action) pairs, then scoring	same	hf
<code>feature_probe</code>	no language model: the same logistic head on the environment's raw numeric features	same	anywhere

`prompt_free` is the "before" condition every other method is measured against. `feature_probe` is the control that answers "does the language model's representation add anything over the raw features?". `random` and `oracle` anchor the score scale; `ucb1` is the classical algorithmic reference on the bandit.

1.2 Tasks All environments are procedurally generated with a seed and have a known oracle, and every rendering contains the full observable state. For the two bandit tasks the oracle is clairvoyant: it knows the hidden arm means or weights, which no policy can read off the text. There it serves as the upper anchor and as the label source for cloning, while UCB1 marks what an algorithm that sees only the history can achieve; the cloned policies learn a history-based approximation of "pull the best arm". They span the decision structures that matter in practice:

Environment	Decision structure	Oracle	Headline metric
<code>bandit</code>	exploration vs exploitation, no state	best true arm	pseudo-regret
<code>contextual_bandit</code>	in-context learning of an abstract rule from (context, action, reward) history	argmax of hidden logistic model	regret
<code>loan</code>	one-shot decisions from semantically meaningful features (tabular, knowledge-driven); OOD switches	approve iff true expected value > 0	profit, regret
<code>gridworld</code>	deterministic multi-step planning	BFS shortest path	success, excess steps
<code>tictactoe</code>	adversarial, perfect information	minimax	win rate
<code>blackjack</code>	risk under uncertainty	basic strategy (hit/stand)	expected return
<code>support</code>	applying a written company policy to a customer's message ("I received the package damaged. I need a refund today.")	the policy's rules	correct-decision rate, payout waste

Every episode return is normalised so that the random policy scores 0 and the oracle scores 1 on that environment; a normalised score is comparable across tasks and the cross-task mean is a single headline number per (model, method).

1.3 Factors

Factor	Levels in the configs	Hypothesis
Parameter count (within one family / recipe)	four ladders run on the laptop, base and instruct at every rung: Qwen2.5 0.5B to 7B, Qwen3 0.6B to 8B, SmoLM2 135M to 1.7B, Gemma 3 270M to 4B (<code>configs/mac_main.yaml</code>); Pythia 70M to 2.8B and OLMo 2 1B and 7B as fully open controls (<code>configs/mac_controls.yaml</code>); the 14B to 72B rungs are future work	H1
Instruction tuning	base vs instruct checkpoint of the same size at every rung of the four ladders	H2
Conversion method	the five methods above	H3, H4
Quantisation	16-bit weights vs Q8_0 and Q4_K_M GGUF served by llama.cpp (generation and first-token scoring only; NF4 needs CUDA and is future work)	H5
Observation / history format	<code>bandit_history_format: summary</code> vs <code>raw</code> ; <code>history_window</code>	H6
Reasoning budget	Qwen3 thinking on/off and chain of thought at a 256-token budget on three tasks (<code>configs/ablations/reasoning.yaml</code>); 2048-token budgets are future work	H7
Prompt details	<code>action_format: name</code> vs <code>letter</code> ; <code>few_shot</code> 0/3; <code>temperature</code> 0.0/0.7; <code>length_norm</code>	secondary
Data budget for adaptation	probe with 10, 40, 100 training episodes; LoRA with 10 and 40 (main sweep: 40, models up to 4B)	H3
Task difficulty	arms 5 vs 10, grid 6 vs 9, opponent random vs minimax, loan shift none / covariate / sign-flip	H1, H3

Notes from the verified literature (`docs/literature_review.md`): the closest prior designs are Jiwatode et al. (2026), who cross Qwen3 sizes with thinking on/off on grid games, and Schmied et al. (ICLR 2026), who find 2B-27B models leave up to 55% of bandit arms untried and that correct rationales often do not translate into the chosen action. H3 has partial prior answers (SFT memorises while RL generalises; a post-trained 3B beating a 72B on Frozen Lake; but diverse-task fine-tuning transfers to unseen tasks), so our contribution on H3 is the controlled, within-family, pre-registered measurement rather than the direction of the effect.

Newer Qwen ladders (Qwen3.5 0.8B-397B, Qwen3.6/3.8) exist and can replace Qwen3 once their repo ids are verified in docs/model_ladders.md. Gemma's large embedding share argues for plotting non-embedding parameters where published (non_embedding_params in the catalogue).

1.4 Pre-registered hypotheses (details and falsification criteria in docs/preregistration.md)

- **H1 (scale).** Under prompt_generate, normalised score increases approximately log-linearly with parameters within a family, with a task-dependent slope: steep for planning and adversarial tasks (gridworld, tictactoe), shallow for one-shot contextual decisions (loan).
- **H2 (instruction tuning).** At equal size, instruct checkpoints beat base checkpoints under prompting; the gap shrinks to non-significance under probe and lora_sft.
- **H3 (adaptation vs scale).** A lora_sft or probe model of at most 1.7B parameters matches or exceeds a zero-shot prompt_generate model ten times larger on in-distribution instances, but not on the out-of-distribution loan variants, where the generalisation gap grows with size.
- **H4 (scoring vs generation).** prompt_score eliminates format failures and closes a substantial fraction (we predict more than half) of the gap between the smallest and largest model under prompt_generate.
- **H5 (quantisation).** Q4_K_M costs less than 0.05 normalised score at 7B and more than 0.10 at 1.5B, under generation and first-token scoring; Q8_0 costs less than 0.05 at every size.
- **H6 (exploration).** On the bandit, no model at any scale reaches the UCB1 reference under prompting; summary history beats raw history for small models.
- **H7 (reasoning).** A 256-token thinking or chain-of-thought budget improves gridworld and tictactoe for models of 4B and above, does not improve or hurts models below 2B, and always loses on latency-normalised efficiency.

2. Phases: where to start, what to do, what each phase produces

Phase 0: harness and sanity (done) This is the part that already exists: six environments with oracles, three backends (hf, openai, mock), five methods, a resumable runner that logs every decision, the statistics (bootstrap confidence intervals, interquartile means, paired permutation tests with Holm correction), and a report generator that writes Markdown and LaTeX tables, figures and an HTML page. The exit criteria are met if the following all pass, and they do:

```
pip install -e ".[hf,dev]" tabulate
pytest -q # unit + integration tests, tiny random model, no downloads
deadeye smoke # full pipeline end-to-end in ~30 s, writes an HTML report
deadeye list-envs; deadeye list-methods
deadeye show-prompt gridworld --seed 3 --cot --few-shot 2 # see exactly what a model sees
```

Phase 1: pilot (week 1) Goal: validate the pipeline on real weights on the Mac, fix prompt wording, measure per-decision latency to size the main sweep, and get first effect sizes. Six checkpoints up to 1.7B, shortened horizons, seeds 5000 to 5019.

```
scripts/run_mac.sh pilot # 3-6 hours; keeps the Mac awake; resumable
deadeye report results/pilot --out report/pilot
deadeye estimate configs/mac_main.yaml --seconds-per-decision 0.4 # re-run with the pilot's measured latency
```

One rule governs the pilot: prompts may be edited here and nowhere else, and only against the pilot seeds. When the pilot is done, fill the measured standard deviations into docs/preregistration.md, tag the repository (git tag prereg-v1) and freeze the design. From that tag on, every run is confirmatory, and any change goes into the deviations table with a date and a reason.

Phase 2: main sweep on the laptop (weeks 2-4) Four ladders, base and instruct at every rung, 100 episodes per cell (300 for tic-tac-toe; blackjack episodes are 20 hands), 100 training episodes for probes; then LoRA (40 training episodes, rungs up to 4B), the free-reply condition on 12 instruct checkpoints, and the control ladders if time allows.

```
3 scripts/run_mac.sh main # ~150-200 laptop-hours, overnight; ladders in the order Qwen2.5, Qwen3, SmoLLM2, Gemma
scripts/run_mac.sh lora # ~50-80 hours
scripts/run_mac.sh free # ~30-40 hours
scripts/run_mac.sh controls # optional, ~30 hours
```

Runs are resumable, so the sweep can be stopped and restarted at will; the order of the ladders means a stopped run still leaves complete ladders behind it, and the paper reports what completed.

Phase 3: ablations and decision models (weeks 5-6) Five ablation blocks at 50 episodes (configs/ablations/: prompting, reasoning, adaptation, tasks with a LoRA companion, and quantisation through llama.cpp), then the decision models that run locally (Kev 0.8B and 4B, Laya, SemIf) against our conversions of the same Qwen3.5 bases.

```
scripts/run_mac.sh ablations
scripts/run_mac.sh quant # after starting the llama.cpp servers listed in the config
scripts/run_mac.sh decision # after starting Kev's server and Laya's package
```

Phase 4: analysis and writing (week 7-9) deadeye report produces every table and figure, including the scale-slope fits (slopes.csv) behind H1 and H4b; deadeye compare runs the paired permutation tests behind each claim, one hypothesis family per call so that the Holm correction is applied within the family:

```
deadeye report results/mac_main results/mac_lora results/mac_free_reply results/mac_controls results/ablations
results/decision_models --out report/paper --paper-dir paper
deadeye compare results/mac_main results/mac_lora \
  --pair "loan:Qwen_Qwen2.5-0.5B-Instruct/lora_sft vs loan:Qwen_Qwen2.5-7B-Instruct/prompt_generate" \
  --pair "gridworld:Qwen_Qwen2.5-0.5B-Instruct/lora_sft vs gridworld:Qwen_Qwen2.5-7B-Instruct/prompt_generate"
``` Fill `paper/sections/*.tex` from the report's `tables/` and `figures/`. The
```

literature review in `docs/literature\_review.md` and the bibliography in `paper/refs.bib` are pre-verified.

#### Phase 5: release (week 9-10)

Tag `v1.0`, build the Docker image, archive the results directory and the report with a DOI (Zenodo), publish the leaderboard HTML, submit the paper (targets: NeurIPS Datasets & Benchmarks, TMLR, COLM, or an agents / small-models workshop), post the preprint.

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### 3. How to test: validity, statistics, and the checks that must pass

#### 3.1 Harness validity (automatic)

- Oracle regret is 0 and the oracle's normalised score is 1 on every environment; the random policy's is 0 (`tests/test\_envs.py`, `tests/test\_metrics\_report.py`).
- Each seed reproduces exactly the same instance and the same pre-drawn noise, so two policies on the same seed face identical potential outcomes (common random numbers; paired tests are valid).
- Exact scoring matches a manual log-probability computation; hidden states differ by layer; LoRA training lowers the loss and raises the probability of the trained label (`tests/test\_hf\_tiny.py`).
- Every prompt is inspectable with `deadeye show-prompt` and every decision is logged with the raw model output, the parsed action, the legal set and the oracle action.

#### 3.2 Statistical protocol (pre-registered)

- Unit of analysis: the episode (seed). Primary outcome: normalised return. Secondary: illegal-action rate, oracle agreement, task metrics (regret, success, win), latency and tokens per decision.
- Point estimates: mean and interquartile mean; uncertainty: 95% bootstrap CI over episodes (2000 resamples). Paired comparisons (same seeds): permutation test on sign flips of the paired differences, 5000 permutations, Holm-corrected within each family of comparisons. Effect sizes: paired Cohen's d.
- Sample size: for a paired comparison,  $n = ((z_{0.975} + z_{0.8}) * sd\_diff / effect)^2$ , implemented as `deadeye.metrics.episodes\_needed(effect, sd)`. With a smallest effect of interest of 0.10, 100 episodes suffice only when the SD of the paired differences is at most 0.36; the pilot measures that SD per task and the pre-registration records it. Tasks whose episode outcome is a single -1/0/+1 game have SDs near 1 in normalised units, which is why tic-tac-toe uses 300 episodes and blackjack packs 20 hands into each of its 100 episodes (the SD of an episode mean falls with the square root of the number of hands). Where even that leaves the detectable effect above 0.10, the paper reports the smallest effect the design could have detected instead of claiming a null.
- Scale curves: fit  $score = a + b * \log_{10}(params)$  per (family, method, task) by least squares with bootstrap CIs on `b`; H1 is supported if  $b > 0$  with CI excluding 0 for `prompt\_generate` and the slopes differ by task class; H4 is supported if `b` under `prompt\_score` is less than half of `b` under `prompt\_generate`.
- Exclusions: a cell is excluded only if the model fails to load or a run crashes; no exclusion on outcome. Every excluded cell is listed in the paper's appendix.

#### 3.3 Leakage and confound controls

- Training seeds (probe, LoRA, few-shot demonstrations) start at 100000 and never overlap evaluation seeds (0-999).
- Prompt wording is frozen after Phase 1 (`prereg-v1` tag). Prompt-format factors are run as explicit ablations, not tuned per model.
- Action labels are compared under both `name` and `letter` encodings so that tokenisation and surface-form effects are measured, not hidden.
- The oracle is deterministic; "oracle agreement" is therefore a lower bound when several actions are optimal (gridworld, tictactoe), which is why return, not agreement, is the primary outcome.
- Quantised models report the catalogue parameter count, not the packed weight count, and are labelled by precision in every table.
- API-served models (`openai` backend) are non-deterministic across servers; results from them are reported in a separate table and never mixed into the scale curves.

#### 3.4 What would make us abandon a hypothesis

Written in `docs/preregistration.md`. In brief: a confidence interval on the relevant slope or difference that includes 0, or an effect in the opposite direction, in the confirmatory sweep.

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### 4. Compute budget

One laptop, no GPU rental. Rough laptop-hours (the pilot replaces these with measured latencies): main sweep 150 to 200, LoRA 50 to 80, free replies 30 to 40, controls about 30, ablations about 100, quantisation block about 15, decision models about 10. Run overnight and resumable, the whole study is five to six weeks of elapsed time. `docs/compute\_budget.md` has the per-size latency table and the decision counts behind these numbers.

### 5. Threats to validity (and what the design does about them)

1. **Prompt sensitivity.** Mitigated by freezing prompts before the sweep and running format ablations; residual sensitivity is reported, not hidden.
2. **Tokenisation and surface-form competition in scoring.** Mitigated by name/letter encodings and

length normalisation as explicit factors.

3. **\*\*Contamination.\*\*** Instances are generated on the fly from seeds; no instance text exists on the web. Rules (blackjack basic strategy, tic-tac-toe) are of course known to models; this is a feature of knowledge-driven decision-making, and the contextual bandit and sign-flipped loan variants are the tasks where prior knowledge cannot help.
4. **\*\*Cross-family confounds.\*\*** Scale claims are made within families only; cross-family plots are descriptive.
5. **\*\*Oracle quality.\*\*** Every oracle is optimal for the configured opponent or rules (expectimax against the random tic-tac-toe opponent, minimax against the minimax opponent, dynamic-programming-optimal hit/stand in blackjack, BFS in gridworld), but blackjack has no doubling or splitting, and oracle agreement is a lower bound wherever several actions are equally good. `docs/known\_limitations.md` keeps the full list.
6. **\*\*Hardware non-determinism.\*\*** Greedy decoding and seeded sampling; the manifest records torch, transformers and the device for every run. Everything is measured on one laptop through one runtime, so latencies are comparable with each other but not with data-centre figures; Metal through PyTorch is also slower than specialised runtimes (MLX, llama.cpp), which is noted wherever speed is reported.
7. **\*\*Scope.\*\*** Weight-level methods stop at 8B and LoRA at 4B because of memory and time on the laptop; the larger rungs are future work, and no claim in the paper extends beyond the sizes that were run.

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### 6. The end state ("done" checklist)

- [ ] `prereg-v1` tag with frozen prompts and `docs/preregistration.md`
- [ ] `results/mac\_main`, `results/mac\_lora`, `results/mac\_free\_reply`, `results/ablations` and `results/decision\_models` complete (no missing cells in each `run\_manifest.json`), or the missing cells listed in the paper's appendix
- [ ] `deadeye` report ... `--paper-dir paper`` regenerates every figure and table in the paper
- [ ] Paper: abstract, introduction, related work (from `docs/literature\_review.md`), method, results (H1-H7 each answered with a number and a CI), limitations, reproducibility statement, appendix with all prompts and the cell list
- [ ] `v1.0` tag, Docker image, Zenodo DOI for code + results, leaderboard HTML published
- [ ] Peer-testing instructions verified by one person who did not write the code (`docs/peer\_testing.md`)

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### 7. The two comparisons the paper makes

The paper's results have two parts, and the second was asked for explicitly: first, small against large open-weight models; second, small open-weight models (converted here) against the purpose-built decision models that appeared in September and October 2026.

#### 7.1 Small versus large open-weight models

Within each family ladder, every method and task: decision quality (normalised score, oracle agreement, task metrics), reliability (illegal-action rate), speed (latency per decision, median and 95th percentile, measured on one machine), efficiency (prompt and generated tokens per decision; an estimate of floating-point operations as 2 x parameters x tokens; peak memory at the recorded precision), and the quality-per-second and quality-per-parameter frontiers. The scale slopes (`slopes.csv`) and the cost frontier (`fig\_pareto`) summarise it; H1, H2, H4, H5 and H7 are the pre-registered tests.

#### 7.2 Small open-weight conversions versus decision models

"Decision models" take a state and typed questions and return probabilities over allowed answers without generating text. The systems and what is verified about them (official cards, announcements and repositories; all performance numbers below are the makers' own):

System	Maker, licence	Base and size	How it decides	How we run it
example	Kev 0.8B / 4B	Jared Palmer; Apache-2.0	Qwen3.5-0.8B/4B-Base (rank-16 LoRA + pointer head)	trained on a 10k-
	Laya / Laya-typed-decisions	Convai Innovations; Apache-2.0	ModernBERT-large 421M encoder + decision head	Choice
	Score / Noul primitives, one forward pass	local package	(adapter if its schema differs)	
	SemIf (formerly OpenJev)	Theo Lee; MIT code	frozen Qwen3.5-4B	reads option logits in one forward pass, no training
	local; and reproduced by our `score_letter`	on the same checkpoint		
	Jev (TypeSafe AI, hosted, closed), Clef-flash 9B / Clef 27B (Cloudflare), pplx-decider 27B (Perplexity), Kev 9B / 27B, Strands Decider 2B (Amazon)			future work: hosted APIs or more memory than the laptop has (`docs/future_work.md`)

On our side: the same Qwen3.5 bases Kev is built on (0.8B, 4B), plus the Qwen2.5 and Qwen3 checkpoints from the main sweep, converted by scoring, probe and LoRA with oracle-labelled states.

Every system is reduced to the same numbers, measured on the same tasks and seeds: normalised score per task and its mean and worst task; oracle agreement; calibration of the probability it assigns to its chosen action against the oracle (expected calibration error and Brier score, which is where "calibrated decisions" is tested rather than assumed); latency per decision (median, 95th percentile, network included for hosted systems and reported separately); tokens and parameters; and openness (weights, code, training data, licence). The report writes this as `system\_comparison.csv` and the paper prints it as one table with the calibration tables beside it.

Two of the rows double as validations. SemIf's readout is our letter-scoring condition, so SemIf on Qwen3.5-4B and our `score\_letter` on the same checkpoint should agree up to prompt wording. Kev on a Qwen3.5 base and our LoRA conversion of the same base differ only in training data (theirs: classification corpora; ours: oracle

labels of the task), which isolates what the training target contributes.

The comparison is reported as exploratory (RQ5) and kept out of the pre-registered tests: the decision models were trained on classification corpora, not on our tasks, so a gap in either direction may reflect training data as much as method. The hosted systems, Jev above all, are future work.

### 8. Future work

`docs/future\_work.md` lists everything the laptop excludes, item by item, with the hardware or service it needs and the approximate cost: the 14B to 72B rungs with every method, the Llama ladder, the larger control rungs, 2048-token reasoning budgets, NF4 quantisation at every size, the hosted decision models and the larger local ones, reinforcement-learning conversions, an MLX or llama.cpp scoring backend, and the ablations at 100 episodes.

#### Extensions after v1.0

- Preference / RL fine-tuning (DPO on oracle-vs-model action pairs, GRPO with environment reward) as a sixth conversion method; the `hf` backend already exposes everything needed.
- Text-game environments (TextWorld / ALFWorld / BabyAI-text) for long-horizon language tasks.
- Multi-agent and negotiation tasks; human-behaviour targets (Centaur-style) instead of oracles.
- Energy measurement per decision (NVML) to replace latency in the Pareto plots.

# Pre-registration

Status: **draft** (to be frozen with the `prereg-v1` git tag after the Phase 1 pilot; any later change is recorded in the "Deviations" section with a date and reason).

### 1. Title

Do small open-weight language models make good decision-makers? A controlled study of scale, conversion method and task structure.

### 2. Research questions

RQ1. How does decision quality scale with parameter count within a model family, and does the slope depend on the task's decision structure?

RQ2. Does the conversion method (prompting, likelihood scoring, hidden-state probe, LoRA behaviour cloning) change the dependence on scale?

RQ3. Which other factors (instruction tuning, quantisation, reasoning budget, observation format, prompt encoding, adaptation data budget) move decision quality, and by how much?

RQ4. What is the quality / inference-cost frontier?

RQ5 (exploratory, added 8 October 2026, not a pre-registered test). How do small open-weight models converted here compare with the purpose-built decision models that run on the same laptop (Kev 0.8B and 4B, Laya, SemIf) in decision quality, calibration, latency and cost on the same tasks and seeds? The hosted decision models (Clef, Perplexity's decider, Jev) are future work.

### 3. Hypotheses, predictions and falsification criteria

Each prediction names the analysis that tests it. "Score" means normalised return (random = 0, oracle = 1). CI means a 95% bootstrap confidence interval. Slopes are from `score = a + b log10(params)` fitted within one family on the sweep results.

#	Prediction	Supported if	Falsified if
H1a	Under `prompt_generate`, `b > 0` in every ladder (SmolLM2 135M-1.7B, Qwen2.5 0.5B-7B, Qwen3 0.6B-8B, Gemma 3 270M-4B)	CI of `b` excludes 0 and is positive for every ladder with $\geq 3$ rungs	any ladder's CI includes 0 or is negative
H1b	Slope is larger for gridworld and tictactoe than for loan	bootstrap CI of (b_planning - b_loan) excludes 0	CI includes 0
H2a	Instruct > base under `prompt_generate` at every size	paired difference CI > 0 at $\geq 80\%$ of sizes	fewer than 50%
H2b	The instruct - base gap under `probe` and `lora_sft` is < 0.05	CI of the mean gap lies within [-0.05, 0.05]	CI excludes that interval
H3a	The smallest rung of each ladder with `lora_sft` $\geq$ the largest rung with `prompt_generate` (ratios 12-15: SmolLM2 135M vs 1.7B, Qwen2.5 0.5B vs 7B, Qwen3 0.6B vs 8B, Gemma 3 270M vs 4B), in distribution	paired difference CI $\geq 0$ on at least 5 of 7 tasks, in at least 3 of 4 ladders	3 or fewer tasks in 2 or more ladders
H3b	On `loan_sign_flip` and `loan_covariate_shift`, the small adapted model's loss relative to in-distribution is larger than the large model's	CI of the interaction excludes 0	CI includes 0
H4a	Illegal-action rate under `prompt_score` is 0	by construction	n/a
H4b	`b_score` < 0.5 `b_generate`	bootstrap CI of the ratio < 0.5	CI includes 0.5 or above
H5	Q4_K_M (llama.cpp GGUF) costs < 0.05 at 7B and > 0.10 at 1.5B under generation and first-token scoring; Q8_0 < 0.05 at every size	CI of the paired 16-bit minus quantised differences meet the thresholds	any fails
H6a	No model under prompting reaches UCB1's bandit score	all CIs below the UCB1 score	any CI includes or exceeds it
H6b	Summary history > raw history for models < 2B	paired CI > 0 for all such models	any CI includes 0
H7a	A 256-token thinking / CoT budget improves gridworld + tictactoe at $\geq 4B$ (Qwen3 4B, 8B; Qwen2.5 7B)	paired CI > 0	CI includes 0
H7b	A 256-token budget does not improve models < 2B (Qwen3 1.7B; Qwen2.5 1.5B)	paired CI includes 0 or is negative	CI > 0
H7c	Latency-normalised score is lower with the budget at every size	score/latency ratio lower for every size	any size higher

### ### 4. Design

- Environments, parameters and seeds exactly as in `configs/mac\_main.yaml`, `configs/mac\_lora.yaml`, `configs/mac\_free\_reply.yaml`, `configs/mac\_controls.yaml`, `configs/decision\_models.yaml` and the files in `configs/ablations/` at the `prereg-v1` tag. Hardware: one Apple M2 Max (32 GB), PyTorch Metal backend, 16-bit weights; LoRA on rungs up to 4B with 40 training episodes (3000 states at most).
- Evaluation seeds 0-99 in the main sweep (tic-tac-toe 0-299; blackjack 0-99 with 20 hands per episode) and 0-49 in the ablations and the free-reply condition (tic-tac-toe 0-149; blackjack 0-49); training seeds 100000-100099; the pilot used seeds 5000-5019 and is not part of the confirmatory data. Prompts frozen at the tag.
- Models: the families and sizes listed in the two configs; a model that cannot be loaded is dropped and listed in the appendix.
- Primary outcome: normalised return per episode. Secondary: illegal-action rate, oracle agreement, task metrics, latency per decision (mean, median, 95th percentile) and tokens per decision, and for every policy that assigns probabilities (scoring, probes, decision models) the expected calibration error and Brier score of the probability assigned to the chosen action against the oracle.

### ### 5. Sample size

For a paired comparison with smallest effect of interest 0.10,  $\alpha = 0.05$  (two-sided) and power 0.8, the required number of paired episodes is  $n = ((1.96 + 0.84) * \text{sd\_diff} / 0.10)^2$ , where  $\text{sd\_diff}$  is the standard deviation of the per-episode paired differences in normalised score. One hundred episodes therefore suffice when  $\text{sd\_diff} \leq 0.36$ . From the pilot (seeds 5000-5019),  $\text{sd\_diff}$  per task (to be filled in at the tag): bandit \_\_, contextual bandit \_\_, loan \_\_, gridworld \_\_, tictactoe \_\_, blackjack \_\_. Tic-tac-toe uses 300 episodes and blackjack 100 episodes of 20 hands because single-game outcomes are -1/0/+1 and their normalised SD is near 1. The ablation blocks and the free-reply condition use 50 episodes, which detect a difference of about 0.14 at  $\text{sd\_diff} = 0.36$  (0.20 at 0.5); that smallest detectable effect is reported next to every ablation result. For any task whose measured  $\text{sd\_diff}$  implies a detectable effect above the stated one, we report it alongside the estimate and do not interpret a non-significant result as evidence of no effect.

### ### 6. Analysis plan

Per cell: mean, IQM, 95% bootstrap CI (2000 resamples). Paired comparisons: permutation test on sign-flipped paired differences (5000 permutations). Holm correction is applied within each hypothesis's family of tests, which is the set of `--pair` arguments of one `deadeye compare` call: H2a is one family per model family, H3a one family per task set, H5 one family per method, H6b and H7a-b one family each. Scale slopes (H1, H4b): ordinary least squares of cell-mean normalised score on  $\log_{10}(\text{parameters})$  within a family, method and task, with 95% CIs from 1000 bootstrap resamples of episodes within cells (`slopes.csv` from `deadeye report`; slope ratios from `slope\_ratio`). All computed by `deadeye report` and `deadeye compare`; the exact code at the `prereg-v1` tag is the analysis.

### ### 7. Exclusion and stopping rules

No episode is excluded for its outcome. A cell is excluded only on load failure or crash (recorded in `run\_manifest.json`). No interim analysis changes the design; the sweep runs to completion.

### ### 8. Deviations (append-only)

Date	What changed	Why
9 October 2026	Scope fixed to one Apple M2 Max laptop: ladders to 8B (LoRA to 4B), GGUF quantisation through llama.cpp, 256-token reasoning budgets, local decision models only, ablations at 50 episodes; everything else moved to future work	the study has no GPU budget; the design was changed before any confirmatory run and before the prereg-v1 tag

### # Testing as a peer

The goal of this page is that someone who did not write the code can (a) verify the harness, (b) reproduce a reported number, (c) evaluate their own model, and (d) submit results.

### ### A. Verify the harness (no downloads, 5 minutes)

```
``bash
git clone https://github.com/CaptiveON/DeadEye && cd DeadEye
python -m venv .venv && source .venv/bin/activate
pip install torch # macOS wheel includes Metal; on Linux add --index-url
https://download.pytorch.org/whl/cpu
pip install -e ".[hf,dev]" tabulate
pytest -q # ~1-2 min: environments, parsing, runner, exact scoring, LoRA, report
deadeye smoke # ~30 s: mock + tiny random model through every env and method
```

deadeye smoke prints the path of an HTML report. Open it: the random row scores 0 and the oracle row scores 1 on every environment; the tiny random model is at chance.

## B. Reproduce a reported number

Every number in the paper comes from a summary.json in a results directory with the config that produced it in run\_manifest.json. To re-run one cell:

```

force
 deadeye run configs/mac_main.yaml --only-env loan --only-model Qwen/Qwen2.5-1.5B-Instruct --only-method prompt_score --
 deadeye report results/mac_main --out report/check
 deadeye compare results/mac_main --pair "loan:Qwen_Qwen2.5-1.5B-Instruct/prompt_score vs loan:Qwen_Qwen2.5-1.5B-
 Instruct/prompt_generate"

```

Greedy decoding and seeded instances make prompt\_score, probe and prompt\_generate (temperature 0) bit-reproducible on the same hardware and library versions, and statistically reproducible across hardware (compare CIs, not point values).

### C. Evaluate your own model

1. **Local weights** (any AutoModelForCausalLM): add to a config

```

models:
 - {backend: hf, id: your-org/your-model, dtype: bfloat16, params: 3000000000, family: yours, instruct: true}

```

params and family are optional for unquantised models (they are counted), required for quantised ones and for the scale plots.

2. **A served model** (vLLM, llama.cpp server, Ollama, LM Studio, OpenRouter):

```

models:
 - {backend: openai, id: served-name, base_url: "http://localhost:8000/v1", params: 7000000000, family: yours}
methods:
 - {name: prompt_generate, params: {max_new_tokens: 16}}
 - {name: prompt_score, params: {action_format: letter}, label: score_letter}

```

Served models support prompt\_generate and first-token prompt\_score; probes and LoRA need local weights.

3. Run `deadeye run your.yaml` and `deadeye report results/yours`.

### D. Add an environment or a method

- Environment: subclass `deadeye.envs.base.Environment`, implement `reset`, `step`, `oracle_action`, `describe`, set `action_labels` and `max_steps`, register it in `deadeye/envs/registry.py`, and add it to `tests/test_envs.py` (the parametrised tests cover legality, determinism and oracle > random automatically).
- Method: subclass `deadeye.policies.base.Policy` (see `policies/probe.py` for an offline-trained example) and register it in `deadeye/policies/registry.py`.

### E. Submit results to the leaderboard

Open a pull request that adds `results/<your-run>/**/summary.json` and `episodes.jsonl` (not `steps.jsonl`, which can be large) plus the config. CI regenerates the leaderboard from all `results/` directories. State hardware, library versions (already in `run_manifest.json`) and whether prompts were modified.

### F. What to look at when something seems wrong

- `steps.jsonl` has the raw model output for every decision; filter `"legal": false` to see format failures and `"parsed_action"` to see what the parser extracted.
- `deadeye show-prompt <env> --seed N --steps K` prints the exact messages.
- `prepare_stats` in `summary.json` reports probe training accuracy and LoRA loss before/after.

## Compute budget

The whole study runs on one laptop through PyTorch's Metal backend at 16-bit precision. The numbers below are estimates for planning; the pilot measures the real per-decision latency of every model (`latency_per_decision_s` in each `summary.json`), and `deadeye estimate <config> --seconds-per-decision <measured>` turns it into hours.

Model size	Scoring or probe (one forward pass)	Generation, 16 tokens	Free reply, 96 tokens
135M to 600M	0.05 to 0.1 s	0.3 to 0.5 s	2 to 3 s
1.5B to 1.7B	0.1 to 0.2 s	0.5 to 0.8 s	3 to 5 s
3B to 4B	0.3 s	1 to 1.5 s	6 to 10 s
7B to 8B	0.6 to 1 s	1.5 to 2.5 s	10 to 15 s

Decisions per model and method in the main sweep are about 17,000 (bandit 5,000; contextual bandit 3,000; loan 2,000; gridworld about 800; tic-tac-toe about 1,200; blackjack about 2,800; support 2,000).

Stage	Config	Rough laptop time
Pilot	<code>configs/pilot.yaml</code>	3 to 6 hours
Main sweep (28 checkpoints, 4 methods)	<code>configs/mac_main.yaml</code>	150 to 200 hours, i.e. two to three weeks of overnight runs
LoRA (24 checkpoints to 4B)	<code>configs/mac_lora.yaml</code>	50 to 80 hours
Free replies (12 checkpoints, 4 tasks, 50 episodes)	<code>configs/mac_free_reply.yaml</code>	30 to 40 hours

Stage	Config	Rough laptop time
Controls (Pythia to 2.8B, OLMo 2 to 7B)	<code>configs/mac_controls.yaml</code>	about 30 hours
Ablations (five blocks plus the LoRA companion, 50 episodes)	<code>configs/ablations/*.yaml</code>	about 100 hours in total
Quantisation block (llama.cpp servers)	<code>configs/ablations/quantisation.yaml</code>	about 15 hours
Decision models (local)	<code>configs/decision_models.yaml</code>	about 10 hours

Everything is resumable, and `scripts/run_mac.sh` runs the stages in order while keeping the Mac awake. Memory: 16-bit weights need about 2 bytes per parameter, so an 8B model takes about 16 GB plus a few GB of working memory; close other applications while the 7B and 8B rungs run.

## Known limitations

Found during the adversarial review of the harness; kept here so the paper's limitations section and the pre-registration stay honest. Items marked *decision* are deliberate and documented rather than bugs.

### Hardware and scope

- **One laptop.** Everything runs on an Apple M2 Max with 32 GB through PyTorch's Metal backend at 16-bit precision. Weight-level methods (exact scoring, probes, LoRA) stop at about 8B parameters and LoRA at 4B; the larger rungs and every hosted system are future work (`docs/future_work.md`). Latencies are comparable within this machine and runtime only; Metal through PyTorch is slower than MLX or llama.cpp.
- **Quantisation across runtimes.** The quantisation block compares 16-bit weights in PyTorch with GGUF Q8\_0 and Q4\_K\_M in llama.cpp, so it varies the runtime as well as the precision, and only generation and first-token scoring apply. *Decision*, forced by the absence of CUDA.
- **Metal was not exercised here.** The backend's device selection is unit-tested, but the sandbox that built the harness had no Apple hardware; the pilot is the first real run on Metal, and `dtype: float16` or `device: cpu` are the fallbacks if `bfloat16` misbehaves.

### Environments

- **Tic-tac-toe oracle.** Against the random opponent the oracle is expectimax (the policy that maximises expected outcome against uniformly random replies); against the minimax opponent it is minimax. The minimax opponent breaks ties at random (seeded), so a deterministic policy meets many distinct games. Oracle agreement remains a lower bound wherever several moves are equally good.
- **Blackjack oracle.** Basic strategy for hit/stand with an infinite deck is the dynamic-programming optimum for these rules (expected value about -0.026 per hand), but the game has no doubling or splitting. Player and dealer cards come from separate pre-drawn streams, so pairing across policies is tight; the variance per hand is still large relative to the random-oracle span, which is why blackjack uses 300 episodes.
- **Contextual bandit and the feature-probe control.** The hidden weights are redrawn every seed and the feature vector is the current context only, so a classifier trained across seeds is at chance by construction. That control therefore measures nothing on this task; the task exists to test in-context learning, which the probe cannot do. *Decision*.
- **Few-shot demonstrations in tic-tac-toe** come from other seeds and may show the agent playing the other mark; each demonstration states its own mark in the observation text. *Decision*.
- **Clairvoyant bandit oracles.** On the bandit and contextual bandit the oracle knows the hidden means or weights, so it is an upper anchor rather than a policy any agent could implement from the text; UCB1 marks the achievable reference, and probes or LoRA trained on oracle labels learn a history-based approximation. *Decision*.
- **Blackjack episodes** can bundle several hands (`hands_per_episode`, 20 in the sweep) so that the per-episode normalised score has a workable variance; each hand is still a separate decision problem and the oracle is unchanged.

### Methods and runner

- **Training caps.** Probes and LoRA stop collecting states at `max_train_states` (default 50000); `prepare_stats` records `n_train_episodes_used` and truncated so a silently smaller data budget is visible in the logs.
- **Out-of-distribution loan tasks.** `train_params` lets adaptation methods train on the in-distribution variant and evaluate on the shifted one; `configs/ablations/tasks.yaml` uses it. Without it, probes and LoRA trained on the shifted task are not out of distribution.
- **Terminating on illegal actions** (`illegal_action: terminate`) cuts episodes short and makes regret look better for early failures; no shipped config uses it. *Decision*.
- **Served models.** An OpenAI-compatible server applies its own sampling defaults, and first-token scoring resolves ties at the floor in favour of the first option. Results from served models are reported separately and never enter the scale curves. *Decision*.

### Decision-model comparison

- **Jev is closed.** Its weights, architecture and training data are not public, so none of the controls used for the open ladders apply to it, and its latency includes a network round trip. The comparison in `configs/decision_models.yaml` is exploratory and stays out of the pre-registered hypotheses. *Decision*.
- **Protocol mapping.** The decision backend sends the task description and observation as the state and the legal actions as the criteria of one choice question, with a fixed instruction; a decision model never sees few-shot demonstrations or a chain-of-thought budget, so only the scoring condition applies to it.
- **Base-model identity.** The open decision models are built on Qwen3.5 and Qwen3.8 checkpoints whose repo ids must be confirmed on the hub before the run; the catalogue does not list them as verified.
- **Protocol adapters.** Kev and Clef speak the System One protocol the decision backend implements. Laya's package and Perplexity's Decisions API have their own schemas and need a thin adapter (same class, different request and response mapping) before they can be run; SemIf is reproduced exactly by `score_letter` on the same checkpoint, so its row can be filled either from its server or from ours.

- **Calibration scope.** ECE and Brier are computed on the probability of the action taken, against the oracle's action; where several actions are optimal the oracle's tie-break makes a well-calibrated policy look over-confident, so calibration is compared across systems within a task, never across tasks.

## Statistics

- Normalisation anchors (random and oracle means) are treated as fixed; their sampling uncertainty is not propagated into the confidence intervals. With 100 episodes it is small relative to the per-cell intervals, and the paired tests do not depend on the anchors at all.
- Holm correction is applied within one report call, which treats every comparison in that report as a single family; the pre-registration names the families per hypothesis and the paper applies the correction per family with `deadeye` `compare`.

## References audit

*Audit date: 8 October 2026. Scope: the 106 entries of `paper/refs.bib` as committed in 57493f6, plus 7 methods-section references added at the coordinator's request. Target style: `natbib` (`authoryear`, `round`, `sort`) with `agsm.bst` (Harvard). Keys were never renamed.*

## Method

- **Primary sources.** Every entry was checked against a publisher or proceedings record: `proceedings.neurips.cc` and `neurips.cc`, `iclr.cc` / `proceedings.iclr.cc` and OpenReview, `proceedings.mlr.press` (ICML, AISTATS, UAI), `aclanthology.org`, `proceedings.mlsys.org`, `ojs.aaai.org`, `colmweb.org`, `link.springer.com`, `nature.com`, PubMed/PMC (PNAS, Nature), `tandfonline.com`, `jstor.org`, `jmlr.org`, `cambridge.org`, `mitpress.mit.edu`, `mheducation.com`, WorldCat; arXiv abstract pages for preprints and technical reports; and the releasing organisation's own pages for model documentation (GitHub organisations QwenLM, meta-llama, huggingface, EleutherAI, AI-Efficiency; `blog.google`; `developers.googleblog.com`; `mistral.ai`; `huggingface.co/blog`). Direct fetches from this container were blocked for most hosts, so confirmation came from search results restricted to those publisher domains and from direct reads of `GitHub/raw.githubusercontent.com`. The page used for each entry is in the table below.
- **Fields confirmed.** Author list (order and spelling), exact title, year, venue, volume/issue/pages where the venue prints them, DOI and arXiv identifier. Where a field could not be confirmed on a primary record it was removed, not guessed: the issue number of Auer et al. (2002), the page ranges of Henderson et al. (2018) and AWQ (MLSys 2024), and all NeurIPS page ranges (NeurIPS shows them only in a BibTeX download that was not reachable). NeurIPS DOIs were added only where the proceedings page itself showed one (Wei et al. 2022; Towers et al. 2025).
- **Published versions.** Every arXiv-only entry was searched for a peer-reviewed version (OpenReview, proceedings and journal sites). Upgraded entries keep the arXiv identifier in `eprint`; where the published title differs from the arXiv title, the arXiv title is kept in `note`.
- **Titles when venue metadata and paper disagree.** For three NeurIPS papers the proceedings index title differs from the title printed on the paper (GTBench: "Capabilities" vs "Limitations"; Hoffmann et al. 2022: "An Empirical Analysis of Compute-Optimal ..." vs "Training Compute-Optimal ..."; Dettmers et al. 2022: "GPT3.int8()" vs "LLM.int8()"). The bib uses the proceedings index title (as NeurIPS's own citation export and DBLP do) and gives the printed/arXiv title in `note`.
- **Author lists.** "and others" was replaced by the full list wherever the list has 30 names or fewer (12 entries). It is kept only where the full list exceeds 30 names: Kadavath et al. (36), Binz et al. 2025 (40), Qwen2.5 (42 named), Qwen3 (60), OLMO 2 (COLM version, Team OLMO + 42), Olmo 3 (67), Llama 3 (559) and the Nature DeepSeek-R1 paper (more than 100). Corporate authors are used where the document is issued by the organisation (Gemma Team, OpenAI, Google, Meta, Mistral AI, Qwen Team).
- **Format for `agsm.bst`.** `agsm` sentence-cases titles, so acronyms and proper nouns are braced; a sentence-case rendering of every title was checked programmatically (0 unprotected acronyms). `agsm` does not print `doi`, `eprint` or `isbn` (kept for the record) and prints `url` through `natbib's \harvardurl`, which typesets it with `\textit` rather than `\url`. A URL containing `_`, `%`, `#`, `&` or `-` would therefore break or corrupt the build, so `url` is used only for web documents (all `@misc` entries) and TMLR papers, all with URLs free of those characters. Web documents carry `howpublished`, `url` and `note = {Accessed 8 October 2026}`. AAAI papers are typed `@article` (journal *Proceedings of the AAAI Conference on Artificial Intelligence*, volume(issue)) because `agsm` warns when an `@inproceedings` has both volume and number.
- **Parser check.** `bibtexparser` 1.4.3 plus a strict custom checker: 113 entries, 113 unique keys, all 106 original keys present, braces balanced in every entry and field, all `agsm`-required fields present for each type, no unescaped `&` `%` `#` `_` in printed fields, page ranges with `--`, no `@` outside entries, `url` fields safe for `\harvardurl`. Result: no problems. (No TeX toolchain is installed, so `bibtex` itself was not run.)
- **Keys used by the paper.** At 22:48 UTC on 8 October the sections cite 73 distinct keys, all present in the bib; the other 40 entries are used by `docs/literature_review.md` or were added for the methods section.

## (a) Summary

	Count
Entries audited (original bib)	106
Kept	106
of which upgraded to a peer-reviewed published version	8
of which corrected (authors, title, venue, year, pages or DOI changed)	38
of which normalised only (full proceedings names, publisher, braces, eprint, url/access date)	59
of which unchanged	1
Removed	0
Added at the coordinator's request (each verified)	7
Entries now in <code>paper/refs.bib</code>	113

Upgraded to the published version: `alain2016probes`; `costarelli2024gamebench`; `deepseekai2025r1`; `liu2025quanthurts`; `olmo2025olmo2`; `tajwar2025curious`; `wang2024lookattext`; `zheng2025qwen3quant`.

Source types (all 113 entries): peer-reviewed (conference): 66; arXiv preprint from established lab: 11; peer-reviewed (journal): 11; official technical report or model card (arXiv): 9; official model documentation (organisation web page / repository): 9; academic book: 4; peer-reviewed workshop paper (non-archival): 2;

official software release (DOI): 1.

## (b) Entry-by-entry audit

"Normalised" means the venue was rewritten to the full proceedings name (e.g. "The Twelfth International Conference on Learning Representations (ICLR 2024)", "Advances in Neural Information Processing Systems 37 (NeurIPS 2024)", "Proceedings of the 42nd International Conference on Machine Learning" with PMLR series/volume), a publisher was added, and title braces were checked; author lists, titles and years in those rows were confirmed unchanged.

Key	Type of source	Venue (as now cited)	Verified at	What changed
abdin2024phi4	official technical report or model card (arXiv)	arXiv preprint arXiv:2412.08905	<a href="https://arxiv.org/abs/2412.08905">https://arxiv.org/abs/2412.08905</a>	Normalised. All 27 authors confirmed; eprint added.
abdin2025phi4reasoning	official technical report or model card (arXiv)	arXiv preprint arXiv:2504.21318	<a href="https://arxiv.org/abs/2504.21318">https://arxiv.org/abs/2504.21318</a>	Corrected. "and others" replaced by the full 23-author list.
agarwal2021precipice	peer-reviewed (conference)	Advances in Neural Information Processing Systems 34 (NeurIPS 2021)	<a href="https://proceedings.neurips.cc/paper/2021/file/f514cec81cb148559cf475e7426eed5e-Paper.pdf">https://proceedings.neurips.cc/paper/2021/file/f514cec81cb148559cf475e7426eed5e-Paper.pdf</a> ; <a href="https://neurips.cc/virtual/2021/poster/26712">https://neurips.cc/virtual/2021/poster/26712</a>	Normalised. Volume moved into booktitle; publisher added.
alain2016probes	peer-reviewed workshop paper (non-archival)	5th International Conference on Learning Representations (ICLR 2017), Workshop Track	<a href="https://openreview.net/forum?id=HJ4-rAVtl">https://openreview.net/forum?id=HJ4-rAVtl</a> ; <a href="https://openreview.net/pdf?id=HJ4-rAVtl">https://openreview.net/pdf?id=HJ4-rAVtl</a>	Upgraded. arXiv preprint -> ICLR 2017 Workshop Track (OpenReview HJ4-rAVtl); year 2016 -> 2017; arXiv id kept.
allal2025smollm2	official technical report or model card (arXiv)	arXiv preprint arXiv:2502.02737	<a href="https://arxiv.org/abs/2502.02737">https://arxiv.org/abs/2502.02737</a>	Corrected. "and others" replaced by the full 22-author list (COLM acceptance not found, kept as arXiv).
auer2002ucb	peer-reviewed (journal)	Machine Learning 47, 235–256	<a href="https://link.springer.com/article/10.1023/A:1013689704352">https://link.springer.com/article/10.1023/A:1013689704352</a>	Corrected. Issue number "2-3" removed (not shown on the Springer record); publisher added.
baldwin1956optimum	peer-reviewed (journal)	Journal of the American Statistical Association 51(275), 429–439	<a href="https://www.tandfonline.com/doi/abs/10.1080/01621459.1956.10501334">https://www.tandfonline.com/doi/abs/10.1080/01621459.1956.10501334</a>	Added. At the coordinator's request; verified (DOI).
belcak2025slm	arXiv preprint from established lab	arXiv preprint arXiv:2506.02153	<a href="https://arxiv.org/abs/2506.02153">https://arxiv.org/abs/2506.02153</a>	Normalised. No published version (NVIDIA Research position paper); eprint added.
belinkov2022probing	peer-reviewed (journal)	Computational Linguistics 48(1), 207–219	<a href="https://aclanthology.org/2022.cl-1.7/">https://aclanthology.org/2022.cl-1.7/</a> ; <a href="https://aclanthology.org/2022.cl-1.7.bib">https://aclanthology.org/2022.cl-1.7.bib</a>	Normalised. Publisher added.
biderman2023pythia	peer-reviewed (conference)	Proceedings of the 40th International Conference on Machine Learning, PMLR 202, pp. 2397–2430	<a href="https://proceedings.mlr.press/v202/biderman23a.html">https://proceedings.mlr.press/v202/biderman23a.html</a>	Corrected. Author name forms as in PMLR (Quentin Gregory Anthony); venue normalised.
biderman2024lessons	arXiv preprint from established lab	arXiv preprint arXiv:2405.14782	<a href="https://arxiv.org/abs/2405.14782">https://arxiv.org/abs/2405.14782</a> ; <a href="https://arxiv.org/html/2405.14782v2">https://arxiv.org/html/2405.14782v2</a>	Corrected. "and others" replaced by the full 30-author v2 list.
binz2023cognitive	peer-reviewed (journal)	Proceedings of the National Academy of Sciences 120(6), e2218523120	<a href="https://pubmed.ncbi.nlm.nih.gov/36730192/">https://pubmed.ncbi.nlm.nih.gov/36730192/</a> ; <a href="https://www.pnas.org/doi/10.1073/pnas.2218523120">https://www.pnas.org/doi/10.1073/pnas.2218523120</a>	Unchanged. All fields confirmed (issue 6 on PMC); no change.
binz2025centaur	peer-reviewed (journal)	Nature 644(8078), 1002–1009	<a href="https://www.nature.com/articles/s41586-025-09215-4">https://www.nature.com/articles/s41586-025-09215-4</a> ; <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12390832/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12390832/</a>	Normalised. 40 authors (>30), so

Key	Type of source	Venue (as now cited)	Verified at	What changed
				leading authors + "and others" kept; note reworded.
brown2020gpt3	peer-reviewed (conference)	Advances in Neural Information Processing Systems 33 (NeurIPS 2020)	<a href="https://proceedings.neurips.cc/paper/2020/file/1457c0d6bfb4967418bfb8ac142f64a-Paper.pdf">https://proceedings.neurips.cc/paper/2020/file/1457c0d6bfb4967418bfb8ac142f64a-Paper.pdf</a> ; <a href="https://neurips.cc/virtual/2020/public/poster_1457c0d6bfb4967418bfb8ac142f64a.html">https://neurips.cc/virtual/2020/public/poster_1457c0d6bfb4967418bfb8ac142f64a.html</a>	Normalised. 31 authors confirmed; volume moved into booktitle; publisher added.
carta2023glam	peer-reviewed (conference)	Proceedings of the 40th International Conference on Machine Learning, PMLR 202, pp. 3676–3713	<a href="https://proceedings.mlr.press/v202/carta23a.html">https://proceedings.mlr.press/v202/carta23a.html</a>	Normalised. Venue normalised.
chen2021decision	peer-reviewed (conference)	Advances in Neural Information Processing Systems 34 (NeurIPS 2021)	<a href="https://proceedings.neurips.cc/paper/2021/hash/7f489f642a0ddb10272b5c3105f0663-Abstract.html">https://proceedings.neurips.cc/paper/2021/hash/7f489f642a0ddb10272b5c3105f0663-Abstract.html</a>	Corrected. Author name as in the NeurIPS record (Misha Laskin); title braces; volume moved into booktitle.
chu2025sft	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 10818–10838	<a href="https://proceedings.mlr.press/v267/chu25c.html">https://proceedings.mlr.press/v267/chu25c.html</a>	Corrected. "and others" replaced by the full 9-author list; pages 10818–10838 added.
costarelli2024gamebench	peer-reviewed workshop paper (non-archival)	Language Gamification Workshop at the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)	<a href="https://neurips.cc/virtual/2024/106557">https://neurips.cc/virtual/2024/106557</a> ; <a href="https://neurips.cc/media/neurips-2024/Slides/106557.pdf">https://neurips.cc/media/neurips-2024/Slides/106557.pdf</a> ; <a href="https://arxiv.org/abs/2406.06613">https://arxiv.org/abs/2406.06613</a>	Upgraded. arXiv preprint -> NeurIPS 2024 Language Gamification workshop paper (peer-reviewed, non-archival); arXiv id kept.
cuadron2025overthinking	arXiv preprint from established lab	arXiv preprint arXiv:2502.08235	<a href="https://arxiv.org/abs/2502.08235">https://arxiv.org/abs/2502.08235</a> ; <a href="https://sky.cs.berkeley.edu/project/agentthink/">https://sky.cs.berkeley.edu/project/agentthink/</a>	Corrected. "and others" replaced by the full 16-author list.
deepseekai2025r1	peer-reviewed (journal)	Nature 645(8081), 633–638	<a href="https://www.nature.com/articles/s41586-025-09422-z">https://www.nature.com/articles/s41586-025-09422-z</a> ; <a href="https://pubmed.ncbi.nlm.nih.gov/40962978/">https://pubmed.ncbi.nlm.nih.gov/40962978/</a>	Upgraded. arXiv preprint -> Nature 645(8081):633–638 (2025), DOI added; author now the Nature byline (Guo et al., >100 names); arXiv id and title kept in note.
dettmers2022llmint8	peer-reviewed (conference)	Advances in Neural Information Processing Systems 35 (NeurIPS 2022)	<a href="https://proceedings.neurips.cc/paper_files/paper/2022/file/c3ba4962c05c49636d4c6206a97e9c8a-Paper-Conference.pdf">https://proceedings.neurips.cc/paper_files/paper/2022/file/c3ba4962c05c49636d4c6206a97e9c8a-Paper-Conference.pdf</a> ; <a href="https://neurips.cc/virtual/2022/poster/53018">https://neurips.cc/virtual/2022/poster/53018</a>	Added. At the coordinator's request; verified. Proceedings index title "GPT3.int8()..." used, printed title "LLM.int8()..." in note.
dettmers2023kbit	peer-reviewed (conference)	Proceedings of the 40th International Conference on Machine Learning, PMLR 202, pp. 7750–7774	<a href="https://proceedings.mlr.press/v202/dettmers23a.html">https://proceedings.mlr.press/v202/dettmers23a.html</a>	Normalised. Venue normalised.
dettmers2023qlora	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	<a href="https://proceedings.neurips.cc/paper_files/paper/2023/hash/1feb87871436031bdc0f2beaa62a049b-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2023/hash/1feb87871436031bdc0f2beaa62a049b-Abstract-Conference.html</a>	Normalised. Volume moved into booktitle; publisher added.
dilkes2025reinforced	arXiv preprint from established lab	arXiv preprint arXiv:2508.10839	<a href="https://arxiv.org/abs/2508.10839">https://arxiv.org/abs/2508.10839</a> ; <a href="https://eprints.soton.ac.uk/505168/">https://eprints.soton.ac.uk/505168/</a>	Normalised. No published version found; eprint added.

Key	Type of source	Venue (as now cited)	Verified at	What changed
duan2024gtbench	peer-reviewed (conference)	Advances in Neural Information Processing Systems 37 (NeurIPS 2024)	<a href="https://proceedings.neurips.cc/paper_files/paper/2024/hash/3191170938b6102e5c203b036b7c16dd-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2024/hash/3191170938b6102e5c203b036b7c16dd-Abstract-Conference.html</a>	Normalised. Volume moved into booktitle (NeurIPS 37); publisher added; note reworded (index title vs PDF/arXiv title).
efron1993bootstrap	academic book	Chapman & Hall (Monographs on Statistics and Applied Probability 57)	<a href="https://search.worldcat.org/title/introduction-to-the-bootstrap/oclc/797437299">https://search.worldcat.org/title/introduction-to-the-bootstrap/oclc/797437299</a> ; <a href="https://openlibrary.org/books/OL1397023M/An_introduction_to_the_bootstrap">https://openlibrary.org/books/OL1397023M/An_introduction_to_the_bootstrap</a> ; <a href="https://www.taylorfrancis.com/books/mono/10.1201/9780429246593/introduction-bootstrap-bradley-efron-tibshirani">https://www.taylorfrancis.com/books/mono/10.1201/9780429246593/introduction-bootstrap-bradley-efron-tibshirani</a>	Added. At the coordinator's request; verified (original Chapman & Hall imprint 1993; T&F DOI of the publisher's listing).
fan2024rational	peer-reviewed (conference)	Proceedings of the AAAI Conference on Artificial Intelligence 38(16), 17960–17967	<a href="https://ojs.aaai.org/index.php/AAAI/article/view/29751">https://ojs.aaai.org/index.php/AAAI/article/view/29751</a>	Corrected. Pages 17960–17967 added; retyped as AAAI journal-style record (38(16)); title braces.
frantar2023gptq	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	<a href="https://iclr.cc/virtual/2023/poster/10855">https://iclr.cc/virtual/2023/poster/10855</a> ; <a href="https://openreview.net/forum?id=tcbBPnfwxS">https://openreview.net/forum?id=tcbBPnfwxS</a>	Normalised. Venue normalised; note reworded.
gao2023lmeval	official software release (DOI)	Zenodo software release of EleutherAI's lm-evaluation-harness, version v0.4.0, 4 December 2023	<a href="https://zenodo.org/records/10256836">https://zenodo.org/records/10256836</a> ; <a href="https://raw.githubusercontent.com/EleutherAI/lm-evaluation-harness/main/CITATION.bib">https://raw.githubusercontent.com/EleutherAI/lm-evaluation-harness/main/CITATION.bib</a>	Normalised. Restructured as web/software record; howpublished, url and access date; non-standard version/publisher fields folded into howpublished.
gema2025inverse	peer-reviewed (journal)	Transactions on Machine Learning Research	<a href="https://openreview.net/pdf/16c9d2ec1a2c68ff255078dd243912d6f9fd1591.pdf">https://openreview.net/pdf/16c9d2ec1a2c68ff255078dd243912d6f9fd1591.pdf</a> ; <a href="https://openreview.net/forum?id=NXgyHW1c7M">https://openreview.net/forum?id=NXgyHW1c7M</a> ; <a href="https://arxiv.org/abs/2507.14417">https://arxiv.org/abs/2507.14417</a>	Corrected. "and others" replaced by the full 14-author list; TMLR OpenReview URL and ISSN added.
gemma2025gemma3	official technical report or model card (arXiv)	arXiv preprint arXiv:2503.19786	<a href="https://arxiv.org/abs/2503.19786">https://arxiv.org/abs/2503.19786</a>	Normalised. Gemma Team corporate author kept (214 contributors); note and eprint added.
google2025gemma3n	official model documentation (organisation web page / repository)	Google Developers Blog, 26 June 2025	<a href="https://developers.googleblog.com/en/introducing-gemma-3n-developer-guide/">https://developers.googleblog.com/en/introducing-gemma-3n-developer-guide/</a> ; <a href="https://ai.google.dev/gemma/docs/releases">https://ai.google.dev/gemma/docs/releases</a>	Normalised. Publication date (26 June 2025) added; url and access date restructured.
google2026gemma4	official model documentation (organisation web page / repository)	Google blog, 2 April 2026 (announcement by C. Farabet and O. Lacombe, Google DeepMind)	<a href="https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/">https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/</a> ; <a href="https://deepmind.google/blog/gemma-4-byte-for-byte-the-most-capable-open-models/">https://deepmind.google/blog/gemma-4-byte-for-byte-the-most-capable-open-models/</a>	Normalised. Publication date (2 April 2026) and byline added; url and access date restructured.
grattafiori2024llama3	official technical report or model card (arXiv)	arXiv preprint arXiv:2407.21783	<a href="https://arxiv.org/abs/2407.21783">https://arxiv.org/abs/2407.21783</a>	Normalised. 559 authors (>30), leading authors + "and others" kept; note shortened; eprint added.
guertler2025textarena	arXiv preprint from established lab	arXiv preprint arXiv:2504.11442	<a href="https://arxiv.org/abs/2504.11442">https://arxiv.org/abs/2504.11442</a>	Normalised. No published version found; arXiv id added as eprint.

Key	Type of source	Venue (as now cited)	Verified at	What changed
guo2017calibration	peer-reviewed (conference)	Proceedings of the 34th International Conference on Machine Learning, PMLR 70, pp. 1321–1330	<a href="https://proceedings.mlr.press/v70/guo17a.html">https://proceedings.mlr.press/v70/guo17a.html</a>	Normalised. Venue normalised.
harris2026explore	peer-reviewed (conference)	Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence, PMLR 337, pp. 2008–2058	<a href="https://proceedings.mlr.press/v337/harris26a.html">https://proceedings.mlr.press/v337/harris26a.html</a>	Corrected. Added pages 2008-2058 (PMLR 337).
hegselmann2023tabllm	peer-reviewed (conference)	Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, PMLR 206, pp. 5549–5581	<a href="https://proceedings.mlr.press/v206/hegselmann23a.html">https://proceedings.mlr.press/v206/hegselmann23a.html</a>	Normalised. Venue normalised; pages confirmed.
henderson2018matters	peer-reviewed (conference)	Proceedings of the AAAI Conference on Artificial Intelligence 32(1)	<a href="https://ojs.aaai.org/index.php/AAAI/article/view/11694">https://ojs.aaai.org/index.php/AAAI/article/view/11694</a>	Corrected. Retyped as AAAI journal-style record (volume 32, issue 1); page range removed (not shown on the AAAI record).
hendrycks2021mmlu	peer-reviewed (conference)	The Ninth International Conference on Learning Representations (ICLR 2021)	<a href="https://openreview.net/forum?id=d7KBjml3GmQ">https://openreview.net/forum?id=d7KBjml3GmQ</a>	Normalised. Venue normalised.
hf2025smollm3	official model documentation (organisation web page / repository)	Hugging Face blog (official release write-up of SmoLLM3)	<a href="https://huggingface.co/blog/smollm3">https://huggingface.co/blog/smollm3</a> ; <a href="https://github.com/huggingface/blog/blob/main/smollm3.md">https://github.com/huggingface/blog/blob/main/smollm3.md</a> ; <a href="https://huggingface.co/HuggingFaceTB/SmolLM3-3B">https://huggingface.co/HuggingFaceTB/SmolLM3-3B</a>	Corrected. Corporate author replaced by the 23 named authors of Hugging Face's own recommended citation; url and access date.
hoffmann2022chinchilla	peer-reviewed (conference)	Advances in Neural Information Processing Systems 35 (NeurIPS 2022)	<a href="https://proceedings.neurips.cc/paper_files/paper/2022/hash/c1e2aff6f588870935f114ebe04a3e5-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2022/hash/c1e2aff6f588870935f114ebe04a3e5-Abstract-Conference.html</a> ; <a href="https://neurips.cc/virtual/2022/poster/53031">https://neurips.cc/virtual/2022/poster/53031</a>	Corrected. Author order and given names set to the NeurIPS proceedings index (Vinyals before Rae; Thomas Hennigan, Katherine Millican, Jack Rae); volume moved into booktitle; note reworded.
holm1979simple	peer-reviewed (journal)	Scandinavian Journal of Statistics 6(2), 65–70	<a href="https://www.jstor.org/stable/i412579">https://www.jstor.org/stable/i412579</a> ; <a href="https://www.jstor.org/stable/4615733">https://www.jstor.org/stable/4615733</a>	Added. At the coordinator's request; verified.
holtzman2021surface	peer-reviewed (conference)	Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, pp. 7038–7051	<a href="https://aclanthology.org/2021.emnlp-main.564/">https://aclanthology.org/2021.emnlp-main.564/</a>	Corrected. DOI, publisher and address added.
hu2022lora	peer-reviewed (conference)	The Tenth International Conference on Learning Representations (ICLR 2022)	<a href="https://openreview.net/forum?id=nZeVKeeFYf9">https://openreview.net/forum?id=nZeVKeeFYf9</a> ; <a href="https://iclr.cc/virtual/2022/poster/6319">https://iclr.cc/virtual/2022/poster/6319</a>	Normalised. Venue normalised.
huang2022zeroshot	peer-reviewed (conference)	Proceedings of the 39th International	<a href="https://proceedings.mlr.press/v162/huang22a.html">https://proceedings.mlr.press/v162/huang22a.html</a>	Corrected. Added PMLR series,

Key	Type of source	Venue (as now cited)	Verified at	What changed
		Conference on Machine Learning, PMLR 162, pp. 9118–9147		volume 162 and pages 9118-9147.
jia2024decision	peer-reviewed (conference)	Advances in Neural Information Processing Systems 37 (NeurIPS 2024)	<a href="https://proceedings.neurips.cc/paper_files/paper/2024/hash/cda04d7ea67ea1376bf8c6962d8541e0-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2024/hash/cda04d7ea67ea1376bf8c6962d8541e0-Abstract-Conference.html</a> ; <a href="https://neurips.cc/virtual/2024/poster/93425">https://neurips.cc/virtual/2024/poster/93425</a>	Normalised. Volume moved into booktitle; publisher added.
jiwatode2026spatial	arXiv preprint from established lab	arXiv preprint arXiv:2607.22732	<a href="https://arxiv.org/abs/2607.22732">https://arxiv.org/abs/2607.22732</a>	Corrected. Unverified "slated for CoG 2026" replaced by the authors' own arXiv comment; no proceedings record yet.
kadavath2022know	arXiv preprint from established lab	arXiv preprint arXiv:2207.05221	<a href="https://arxiv.org/abs/2207.05221">https://arxiv.org/abs/2207.05221</a>	Normalised. 36 authors (>30), so leading authors + "and others" kept; eprint added.
kaplan2020scaling	arXiv preprint from established lab	arXiv preprint arXiv:2001.08361	<a href="https://arxiv.org/abs/2001.08361">https://arxiv.org/abs/2001.08361</a>	Normalised. No published version (OpenAI); eprint added.
krishnamurthy2024explore	peer-reviewed (conference)	Advances in Neural Information Processing Systems 37 (NeurIPS 2024)	<a href="https://proceedings.neurips.cc/paper_files/paper/2024/hash/d951f73c521d069fefbb73396df01424-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2024/hash/d951f73c521d069fefbb73396df01424-Abstract-Conference.html</a>	Normalised. Volume moved into booktitle; publisher added.
kumar2025precision	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	<a href="https://iclr.cc/virtual/2025/oral/31737">https://iclr.cc/virtual/2025/oral/31737</a> ; <a href="https://iclr.cc/virtual/2025/poster/27832">https://iclr.cc/virtual/2025/poster/27832</a>	Normalised. Venue normalised.
laskin2023incontext	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	<a href="https://openreview.net/pdf?id=hy0a5MMPUy">https://openreview.net/pdf?id=hy0a5MMPUy</a> ; <a href="https://iclr.cc/virtual/2023/poster/11148">https://iclr.cc/virtual/2023/poster/11148</a>	Normalised. Venue normalised.
lattimore2020bandit	academic book	Cambridge University Press	<a href="https://www.cambridge.org/9781108486828">https://www.cambridge.org/9781108486828</a> ; <a href="https://www.cambridge.org/core/books/bandit-algorithms/8E39FD004F6CE036680F90DD0C6F09FC">https://www.cambridge.org/core/books/bandit-algorithms/8E39FD004F6CE036680F90DD0C6F09FC</a>	Normalised. Address added.
law2015simulation	academic book	McGraw-Hill Education, Fifth edn	<a href="https://www.mheducation.com/highered/product/simulation-modeling-and-analysis-law.html">https://www.mheducation.com/highered/product/simulation-modeling-and-analysis-law.html</a>	Added. At the coordinator's request (preferred common-random-numbers reference); verified; Glasserman and Yao (1992) not added.
li2025smallstruggle	peer-reviewed (conference)	Findings of the Association for Computational Linguistics: ACL 2025, pp. 25366–25394	<a href="https://aclanthology.org/2025.findings-acl.1301/">https://aclanthology.org/2025.findings-acl.1301/</a>	Corrected. "and others" replaced by the full 8-author list; pages 25366-25394, DOI, publisher and address added.
lin2024awq	peer-reviewed (conference)	Proceedings of Machine Learning and Systems 6 (MLSys 2024)	<a href="https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html">https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html</a>	Normalised. Venue normalised (MLSys 2024, vol. 6); page range not added (seen only in a citing paper).
lin2024urial	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://proceedings.iclr.cc/paper_files/paper/2024/hash/6bcbb4a501dbad0eba1b660c1a55318c-Abstract-Conference.html">https://proceedings.iclr.cc/paper_files/paper/2024/hash/6bcbb4a501dbad0eba1b660c1a55318c-Abstract-Conference.html</a>	Normalised. Venue normalised.

Key	Type of source	Venue (as now cited)	Verified at	What changed
liu2024agentbench	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://proceedings.iclr.cc/paper_files/paper/2024/file/e9df36b21ff4ee211a8b71ee8b7e9f57-Paper-Conference.pdf">https://proceedings.iclr.cc/paper_files/paper/2024/file/e9df36b21ff4ee211a8b71ee8b7e9f57-Paper-Conference.pdf</a> ; <a href="https://iclr.cc/virtual/2024/poster/17388">https://iclr.cc/virtual/2024/poster/17388</a>	Normalised. All 22 authors confirmed; venue normalised.
liu2025mindstep	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 38489–38517	<a href="https://proceedings.mlr.press/v267/liu25t.html">https://proceedings.mlr.press/v267/liu25t.html</a>	Normalised. Venue normalised.
liu2025quanthurts	peer-reviewed (conference)	Second Conference on Language Modeling (COLM 2025)	<a href="https://colmweb.org/2025/AcceptedPapers.html">https://colmweb.org/2025/AcceptedPapers.html</a> ; <a href="https://openreview.net/forum?id=BM192Ps5Nv">https://openreview.net/forum?id=BM192Ps5Nv</a>	Upgraded. arXiv preprint -> COLM 2025 (OpenReview BM192Ps5Nv); arXiv id kept.
liu2025rational	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	<a href="https://proceedings.iclr.cc/paper_files/paper/2025/hash/7cdb411d07bdf6cd43ddf1254fae98cd-Abstract-Conference.html">https://proceedings.iclr.cc/paper_files/paper/2025/hash/7cdb411d07bdf6cd43ddf1254fae98cd-Abstract-Conference.html</a> ; <a href="https://openreview.net/forum?id=dAeET8gxqg">https://openreview.net/forum?id=dAeET8gxqg</a>	Normalised. Venue normalised.
liu2026spiral	peer-reviewed (conference)	The Fourteenth International Conference on Learning Representations (ICLR 2026)	<a href="https://iclr.cc/virtual/2026/poster/10011289">https://iclr.cc/virtual/2026/poster/10011289</a> ; <a href="https://proceedings.iclr.cc/paper_files/paper/2026/file/3ba7c8c36b71b2fc31fae3361aefefbf-Paper-Conference.pdf">https://proceedings.iclr.cc/paper_files/paper/2026/file/3ba7c8c36b71b2fc31fae3361aefefbf-Paper-Conference.pdf</a>	Corrected. Author order corrected to the ICLR 2026 camera-ready (Yu second, Guertler fourth).
meta2025llamamodels	official model documentation (organisation web page / repository)	Official GitHub repository meta-llama/llama-models (model cards, release table and licences for Llama 3.1, 3.2, 3.3 and 4)	<a href="https://github.com/meta-llama/llama-models">https://github.com/meta-llama/llama-models</a>	Corrected. Invented descriptive title replaced by the repository title ("Llama Models"); url and access date restructured.
mistral2025mistral3	official model documentation (organisation web page / repository)	Mistral AI news announcement, 2 December 2025 (Mistral Large 3 and the Ministral 3 family)	<a href="https://mistral.ai/news/mistral-3/">https://mistral.ai/news/mistral-3/</a>	Corrected. Title corrected to the page title "Introducing Mistral 3"; url and access date restructured.
mistral2025small3	official model documentation (organisation web page / repository)	Mistral AI news announcement, 30 January 2025	<a href="https://mistral.ai/news/mistral-small-3/">https://mistral.ai/news/mistral-small-3/</a>	Normalised. Title braces; date, url and access date restructured.
mistral2025small31	official model documentation (organisation web page / repository)	Mistral AI news announcement, 17 March 2025	<a href="https://mistral.ai/news/mistral-small-3-1/">https://mistral.ai/news/mistral-small-3-1/</a>	Normalised. Title braces; date, url and access date restructured.
monea2025icrl	peer-reviewed (conference)	Second Conference on Language Modeling (COLM 2025)	<a href="https://openreview.net/forum?id=c0RsezY2D1">https://openreview.net/forum?id=c0RsezY2D1</a>	Normalised. COLM 2025 confirmed on OpenReview; venue normalised.
nie2025evolve	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 46346–46376	<a href="https://proceedings.mlr.press/v267/nie25b.html">https://proceedings.mlr.press/v267/nie25b.html</a>	Corrected. Author name form as in PMLR (Jonathan Lee); venue normalised.
olmo2025olmo2	peer-reviewed (conference)	Second Conference on Language Modeling (COLM 2025)	<a href="https://openreview.net/pdf?id=2ezugTT9kU">https://openreview.net/pdf?id=2ezugTT9kU</a> ; <a href="https://colmweb.org/2025/AcceptedPapers.html">https://colmweb.org/2025/AcceptedPapers.html</a> ; <a href="https://arxiv.org/abs/2501.00656">https://arxiv.org/abs/2501.00656</a>	Upgraded. arXiv preprint -> COLM 2025 "2 OLMo 2 Furious (COLM's Version)" (OpenReview 2ezugTT9kU); author list as in

Key	Type of source	Venue (as now cited)	Verified at	What changed
				COLM (Walsh et al., >30); arXiv id kept.
olmo2025olmo3	official technical report or model card (arXiv)	arXiv preprint arXiv:2512.13961	<a href="https://arxiv.org/abs/2512.13961">https://arxiv.org/abs/2512.13961</a>	Corrected. Author list truncated to the confirmed leading entries (Team Olmo, Ettinger; 67 names, >30); eprint added.
openai2025gptoss	official technical report or model card (arXiv)	arXiv preprint arXiv:2508.10925	<a href="https://arxiv.org/abs/2508.10925">https://arxiv.org/abs/2508.10925</a>	Normalised. OpenAI corporate author kept (124 named contributors); eprint added.
orgad2025know	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	<a href="https://proceedings.iclr.cc/paper_files/paper/2025/hash/a712d461e57201efe35d429a6f1731c1-Abstract-Conference.html">https://proceedings.iclr.cc/paper_files/paper/2025/hash/a712d461e57201efe35d429a6f1731c1-Abstract-Conference.html</a>	Corrected. "and others" replaced by the full 7-author list.
paglieri2025balrog	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	<a href="https://iclr.cc/virtual/2025/poster/28856">https://iclr.cc/virtual/2025/poster/28856</a> ; <a href="https://proceedings.iclr.cc/paper_files/paper/2025/file/f0b1515be276f6ba82b4f2b25e50bef0-Paper-Conference.pdf">https://proceedings.iclr.cc/paper_files/paper/2025/file/f0b1515be276f6ba82b4f2b25e50bef0-Paper-Conference.pdf</a>	Corrected. Author spelling Wolczyk -> Wolczyk; Foerster given name as in the ICLR record; venue normalised.
pedregosa2011scikit	peer-reviewed (journal)	Journal of Machine Learning Research 12(85), 2825–2830	<a href="https://jmlr.org/papers/v12/pedregosa11a.html">https://jmlr.org/papers/v12/pedregosa11a.html</a>	Added. At the coordinator's request; verified (16 authors).
pomerleau1988alvinn	peer-reviewed (conference)	Advances in Neural Information Processing Systems 1 (NIPS 1988), pp. 305–313	<a href="https://proceedings.neurips.cc/paper/1988/file/812b4ba287f5ee0bc9d43bbf5bbe87fb-Paper.pdf">https://proceedings.neurips.cc/paper/1988/file/812b4ba287f5ee0bc9d43bbf5bbe87fb-Paper.pdf</a> ; <a href="https://proceedings.neurips.cc/paper/1988">https://proceedings.neurips.cc/paper/1988</a> ; <a href="https://www.ri.cmu.edu/publications/alvinn-an-autonomous-land-vehicle-in-a-neural-network/">https://www.ri.cmu.edu/publications/alvinn-an-autonomous-land-vehicle-in-a-neural-network/</a>	Corrected. Editor (D. S. Touretzky) and publisher (Morgan Kaufmann) added; pages confirmed; note on the 1989 volume date.
qwen2024qwen25	official technical report or model card (arXiv)	arXiv preprint arXiv:2412.15115	<a href="https://arxiv.org/abs/2412.15115">https://arxiv.org/abs/2412.15115</a> ; <a href="https://github.com/QwenLM/Qwen3">https://github.com/QwenLM/Qwen3</a>	Corrected. Author form follows the Qwen team's own citation (An Yang et al.; 42 named, >30); group name moved to note; eprint added.
qwen2025qwen3github	official model documentation (organisation web page / repository)	Official GitHub repository QwenLM/Qwen3 (model release notes, including the Qwen3-2507 updates)	<a href="https://github.com/QwenLM/Qwen3">https://github.com/QwenLM/Qwen3</a>	Corrected. Invented descriptive title replaced by the repository title ("Qwen3"); url and access date restructured.
qwen2026github	official model documentation (organisation web page / repository)	Official GitHub repository QwenLM/Qwen3.8 (release notes for Qwen3.5, Qwen3.6 and Qwen3.8)	<a href="https://github.com/QwenLM/Qwen3.8">https://github.com/QwenLM/Qwen3.8</a>	Corrected. Invented descriptive title replaced by the repository title ("Qwen3.8"); url and access date restructured.
rafaailov2023dpo	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	<a href="https://papers.nips.cc/paper/2023/hash/a85b405ed65c6477a4fe8302b5e06ce7-Abstract-Conference.html">https://papers.nips.cc/paper/2023/hash/a85b405ed65c6477a4fe8302b5e06ce7-Abstract-Conference.html</a> ; <a href="https://neurips.cc/virtual/2023/poster/72164">https://neurips.cc/virtual/2023/poster/72164</a>	Normalised. Order Manning before Ermon confirmed on the NeurIPS abstract page; note removed.
robinson2023mcsb	peer-reviewed (conference)	The Eleventh International Conference on	<a href="https://iclr.cc/virtual/2023/poster/10737">https://iclr.cc/virtual/2023/poster/10737</a> ; <a href="https://openreview.net/forum?id=vKbprrajc5B">https://openreview.net/forum?id=vKbprrajc5B</a>	Corrected. Author list set to the ICLR 2023 record

Key	Type of source	Venue (as now cited)	Verified at	What changed
		Learning Representations (ICLR 2023)		(Robinson and Wingate); arXiv lists Rytting as well.
ross2011dagger	peer-reviewed (conference)	Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, PMLR 15, pp. 627–635	<a href="https://proceedings.mlr.press/v15/ross11a.html">https://proceedings.mlr.press/v15/ross11a.html</a>	Corrected. PMLR series/volume 15 and address added; author name forms as in PMLR.
ruoss2025lmact	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 52303–52344	<a href="https://proceedings.mlr.press/v267/ruoss25a.html">https://proceedings.mlr.press/v267/ruoss25a.html</a>	Normalised. Confirmed PMLR 267:52303–52344; venue normalised, publisher added.
schaeffer2023mirage	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	<a href="https://proceedings.neurips.cc/paper_files/paper/2023/hash/ad98a266f45005c403b8311ca7e8bd7-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2023/hash/ad98a266f45005c403b8311ca7e8bd7-Abstract-Conference.html</a>	Normalised. Volume moved into booktitle; publisher added.
schmied2026greedy	peer-reviewed (conference)	The Fourteenth International Conference on Learning Representations (ICLR 2026)	<a href="https://proceedings.iclr.cc/paper_files/paper/2026/hash/cf2fcd40de8520027af2eadcafdec6-Abstract-Conference.html">https://proceedings.iclr.cc/paper_files/paper/2026/hash/cf2fcd40de8520027af2eadcafdec6-Abstract-Conference.html</a>	Normalised. Venue normalised (ICLR 2026).
sclar2024formatting	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html">https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html</a>	Normalised. Venue normalised.
shao2024deepseekmath	arXiv preprint from established lab	arXiv preprint arXiv:2402.03300	<a href="https://arxiv.org/abs/2402.03300">https://arxiv.org/abs/2402.03300</a>	Normalised. No published version (DeepSeek-AI); GRPO note removed; eprint added.
shinn2023reflexion	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	<a href="https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html</a>	Corrected. Author list set to the NeurIPS 2023 record (5 authors); Edward Berman appears only on arXiv v4. Note removed.
shojaee2025illusion	peer-reviewed (conference)	Advances in Neural Information Processing Systems 38 (NeurIPS 2025)	<a href="https://proceedings.neurips.cc/paper_files/paper/2025/hash/9b26ad15462c81548c0689188d2e8018-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2025/hash/9b26ad15462c81548c0689188d2e8018-Abstract-Conference.html</a> ; <a href="https://openreview.net/forum?id=YghiOusmvw">https://openreview.net/forum?id=YghiOusmvw</a>	Normalised. Volume (NeurIPS 38) added to booktitle; publisher added.
sinha2026illusion	peer-reviewed (conference)	The Fourteenth International Conference on Learning Representations (ICLR 2026)	<a href="https://iclr.cc/virtual/2026/poster/10011614">https://iclr.cc/virtual/2026/poster/10011614</a>	Normalised. Venue normalised.
skean2025layer	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 55854–55875	<a href="https://proceedings.mlr.press/v267/skean25a.html">https://proceedings.mlr.press/v267/skean25a.html</a>	Corrected. "and others" replaced by the full 7-author list.
snell2025ttc	peer-reviewed (conference)	The Thirteenth International Conference on Learning Representations (ICLR 2025)	<a href="https://proceedings.iclr.cc/paper_files/paper/2025/file/1b623663fd9b874366f3ce019fdffd44-Paper-Conference.pdf">https://proceedings.iclr.cc/paper_files/paper/2025/file/1b623663fd9b874366f3ce019fdffd44-Paper-Conference.pdf</a> ; <a href="https://iclr.cc/virtual/2025/oral/31924">https://iclr.cc/virtual/2025/oral/31924</a>	Corrected. Title changed to the ICLR 2025 title ("... than Scaling Parameters for Reasoning"); arXiv title moved to note.
sutton2018rl	academic book	MIT Press, Second edn	<a href="https://mitpress.mit.edu/9780262039246/reinforcement-learning/">https://mitpress.mit.edu/9780262039246/reinforcement-learning/</a>	Normalised. Address added.

Key	Type of source	Venue (as now cited)	Verified at	What changed
szot2024llarp	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://proceedings.iclr.cc/paper_files/paper/2024/file/8f9461d11e8ce9b0b28b21bfc645d74e-Paper-Conference.pdf">https://proceedings.iclr.cc/paper_files/paper/2024/file/8f9461d11e8ce9b0b28b21bfc645d74e-Paper-Conference.pdf</a> ; <a href="https://openreview.net/pdf?id=u6imHU4Ebu">https://openreview.net/pdf?id=u6imHU4Ebu</a>	Normalised. Venue normalised.
tajwar2025curious	peer-reviewed (conference)	Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, pp. 58227–58281	<a href="https://proceedings.mlr.press/v267/tajwar25a.html">https://proceedings.mlr.press/v267/tajwar25a.html</a> ; <a href="https://icml.cc/virtual/2025/poster/45106">https://icml.cc/virtual/2025/poster/45106</a>	Upgraded. arXiv preprint -> ICML 2025, PMLR 267:58227–58281; arXiv id kept.
tam2024speakfreely	peer-reviewed (conference)	Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track, pp. 1218–1236	<a href="https://aclanthology.org/2024.emnlp-industry.91/">https://aclanthology.org/2024.emnlp-industry.91/</a>	Normalised. Address and publisher added; arXiv title recorded in note.
tan2024twosome	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://openreview.net/forum?id=hILVmJ4Uvu">https://openreview.net/forum?id=hILVmJ4Uvu</a>	Normalised. Venue normalised.
towers2025gymnasium	peer-reviewed (conference)	Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets and Benchmarks Track	<a href="https://proceedings.neurips.cc/paper_files/paper/2025/hash/d7ff1795e8527f6443371c3933bdb52b-Abstract-Datasets_and_Benchmarks_Track.html">https://proceedings.neurips.cc/paper_files/paper/2025/hash/d7ff1795e8527f6443371c3933bdb52b-Abstract-Datasets_and_Benchmarks_Track.html</a> ; <a href="https://neurips.cc/virtual/2025/poster/121446">https://neurips.cc/virtual/2025/poster/121446</a> ; <a href="https://github.com/Farama-Foundation/Gymnasium">https://github.com/Farama-Foundation/Gymnasium</a>	Normalised. NeurIPS 38 D&B and DOI confirmed; title braces; publisher added.
wang2023selfconsistency	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	<a href="https://openreview.net/forum?id=1PL1NIMMrw">https://openreview.net/forum?id=1PL1NIMMrw</a> ; <a href="https://iclr.cc/virtual/2023/poster/11718">https://iclr.cc/virtual/2023/poster/11718</a>	Normalised. Quoc V. Le initial; venue normalised.
wang2024lookattext	peer-reviewed (conference)	First Conference on Language Modeling (COLM 2024)	<a href="https://openreview.net/forum?id=qHdSA85GyZ">https://openreview.net/forum?id=qHdSA85GyZ</a> ; <a href="https://colmweb.org/2024/AcceptedPapers.html">https://colmweb.org/2024/AcceptedPapers.html</a>	Upgraded. arXiv preprint -> COLM 2024 (OpenReview qHdSA85GyZ); arXiv id kept.
wang2024myanswer	peer-reviewed (conference)	Findings of the Association for Computational Linguistics: ACL 2024, pp. 7407–7416	<a href="https://aclanthology.org/2024.findings-acl.441/">https://aclanthology.org/2024.findings-acl.441/</a>	Corrected. "and others" replaced by the full 8-author list; DOI, publisher and address added.
wang2025ragen	arXiv preprint from established lab	arXiv preprint arXiv:2504.20073	<a href="https://arxiv.org/abs/2504.20073">https://arxiv.org/abs/2504.20073</a>	Corrected. "and others" replaced by the full 18-author list.
wei2022cot	peer-reviewed (conference)	Advances in Neural Information Processing Systems 35 (NeurIPS 2022)	<a href="https://proceedings.neurips.cc/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html">https://proceedings.neurips.cc/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html</a>	Corrected. DOI 10.52202/068431-1800 added; Ed H. Chi as in the paper.
wei2022emergent	peer-reviewed (journal)	Transactions on Machine Learning Research	<a href="https://openreview.net/forum?id=yzkSU5zdWd">https://openreview.net/forum?id=yzkSU5zdWd</a> ; <a href="https://arxiv.org/pdf/2206.07682">https://arxiv.org/pdf/2206.07682</a>	Normalised. TMLR OpenReview URL and ISSN added; journal name normalised.
wolf2020transformers	peer-reviewed (conference)	Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp. 38–45	<a href="https://aclanthology.org/2020.emnlp-demos.6/">https://aclanthology.org/2020.emnlp-demos.6/</a>	Added. At the coordinator's request; verified (22 authors, DOI).

Key	Type of source	Venue (as now cited)	Verified at	What changed
wu2024smartplay	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://proceedings.iclr.cc/paper_files/paper/2024/hash/063d1fadbd557466bf5bcffe814a64e-Abstract-Conference.html">https://proceedings.iclr.cc/paper_files/paper/2024/hash/063d1fadbd557466bf5bcffe814a64e-Abstract-Conference.html</a>	Normalised. Venue normalised to full ICLR 2024 name; publisher added.
yang2025qwen3	official technical report or model card (arXiv)	arXiv preprint arXiv:2505.09388	<a href="https://arxiv.org/abs/2505.09388">https://arxiv.org/abs/2505.09388</a> ; <a href="https://github.com/QwenLM/Qwen3">https://github.com/QwenLM/Qwen3</a>	Normalised. 60 authors (>30), leading authors + "and others" kept; note and eprint added.
yao2023react	peer-reviewed (conference)	The Eleventh International Conference on Learning Representations (ICLR 2023)	<a href="https://openreview.net/forum?id=WE_vluYUL-X">https://openreview.net/forum?id=WE_vluYUL-X</a> ; <a href="https://iclr.cc/virtual/2023/oral/12647">https://iclr.cc/virtual/2023/oral/12647</a>	Normalised. Venue normalised; publisher added.
zhao2021calibrate	peer-reviewed (conference)	Proceedings of the 38th International Conference on Machine Learning, PMLR 139, pp. 12697–12706	<a href="https://proceedings.mlr.press/v139/zhao21c/zhao21c.pdf">https://proceedings.mlr.press/v139/zhao21c/zhao21c.pdf</a>	Normalised. Venue normalised; internal note (PMLR lists "Zihao Zhao") moved to this audit.
zheng2024selectors	peer-reviewed (conference)	The Twelfth International Conference on Learning Representations (ICLR 2024)	<a href="https://openreview.net/pdf?id=shr9PXz7T0">https://openreview.net/pdf?id=shr9PXz7T0</a>	Normalised. Venue normalised.
zheng2025qwen3quant	peer-reviewed (journal)	Visual Intelligence 4, Article 11	<a href="https://link.springer.com/article/10.1007/s44267-026-00114-4">https://link.springer.com/article/10.1007/s44267-026-00114-4</a> ; <a href="https://github.com/AI-Efficiency/Qwen3-Quantization-Toolkit">https://github.com/AI-Efficiency/Qwen3-Quantization-Toolkit</a>	Upgraded. arXiv preprint -> Visual Intelligence 4, article 11 (Springer, 2026), DOI added; journal author list adds Zining Wang; year 2025 -> 2026; arXiv id kept.
zhou2023ifeval	arXiv preprint from established lab	arXiv preprint arXiv:2311.07911	<a href="https://arxiv.org/abs/2311.07911">https://arxiv.org/abs/2311.07911</a>	Normalised. No published version (Google); eprint added.
zhou2023lima	peer-reviewed (conference)	Advances in Neural Information Processing Systems 36 (NeurIPS 2023)	<a href="https://proceedings.neurips.cc/paper_files/paper/2023/hash/ac662d74829e4407ce1d126477f4a03a-Abstract-Conference.html">https://proceedings.neurips.cc/paper_files/paper/2023/hash/ac662d74829e4407ce1d126477f4a03a-Abstract-Conference.html</a>	Corrected. "and others" replaced by the full 15-author list.

### (c) Removed

**None.** Every one of the 106 original entries was verified against a primary source, and every entry meets the source rule. 72 are peer-reviewed conference or journal papers (64 conference, 8 journal). 2 are peer-reviewed workshop papers. 2 are academic books. 11 are arXiv preprints from established groups (OpenAI, Anthropic, Google, NVIDIA Research, DeepSeek-AI, EleutherAI, UC Berkeley, Northwestern/Stanford/Microsoft, A\*STAR/NUS/MIT-IBM, University of Southampton, Leibniz University Hannover) with no published version. 9 are technical reports or model cards on arXiv from the releasing organisation. 9 are official documentation pages or repositories of the releasing organisation, and 1 is a DOI-registered software release by its maintainers. Because nothing was removed, no removal search of paper/sections/\*.tex or docs/literature\_review.md was needed.

Two entries are closest to the line. If the owner reads the rule more strictly, these are the candidates, with where they are used (22:48 UTC) and a replacement already in the bib:

Key	Why borderline	Used in	Suggested replacement
costarelli2024gamebench	Peer-reviewed but non-archival NeurIPS 2024 workshop paper by a multi-university student team (Olin, UT Dallas, Columbia, Oxford, UCL, ShanghaiTech, Manchester, two independent), not an established lab	docs/literature_review.md §2.1 (annotation, original line 79); not cited in paper/sections/*.tex at audit time	duan2024gtbench (NeurIPS 2024 main track game-theoretic benchmark)
gao2023lmeval	Software release on Zenodo (DOI, maintainers' recommended citation), not a paper or model documentation	docs/literature_review.md §2.7 (annotation, original line 414); not cited in paper/sections/*.tex at audit time	biderman2024lessons (EleutherAI paper describing lm-evaluation-harness)

#### (d) Entries kept with residual doubt

- `robinson2023mcsb`: the ICLR 2023 record (`iclr.cc` poster page, OpenReview `yKbprarjc5B`) lists Joshua Robinson and David Wingate. `arXiv:2210.12353` also lists Christopher Michael Rytting. The bib follows the venue record. If the camera-ready PDF on OpenReview shows three authors, add Rytting back as the second author.
- `shinn2023reflexion`: the NeurIPS 2023 proceedings page lists five authors. `arXiv v4` (10 Oct 2023) adds Edward Berman. The bib follows the proceedings.
- `hoffmann2022chinchilla`, `duan2024gtbench`, `dettmers2022llmint8`: proceedings index title and printed paper title differ (see Method). For Hoffmann et al., the NeurIPS index also orders Vinyals before Rae and uses the given names Thomas, Katherine and Jack, while `arXiv` has Rae before Vinyals. The bib follows the index for both title and authors, and gives the printed title in note.
- `towers2025gymnasium`: the `arXiv` record, the Farama README and (per search) the proceedings PDF put Jordan Terry third, but the NeurIPS virtual-site metadata lists J. K. Terry later. The paper order is kept.
- `jiwatode2026spatial`: `arXiv`-only (22 July 2026). The authors' `arXiv` comment says "To be published at COG 2026" (IEEE CoG, Madrid, 1-4 September 2026), but no proceedings or IEEE Xplore record was found yet. Replace with the CoG record when it appears. The review's Section 5 calls it "(CoG 2026)". That prose was not edited.
- `dilkes2025reinforced`: a single-version `arXiv` preprint (14 August 2025) from the University of Southampton. It is acceptable as an academic-lab preprint, but it is not peer reviewed and the review leans on it for H3 (the 3B > 72B Frozen Lake result).
- `costarelli2024gamebench` and `gao2023lmeval`: see the Removed section above.
- `zheng2025qwen3quant`: the journal version (Visual Intelligence 4, article 11, 16 April 2026; DOI confirmed on Springer) adds Zining Wang. The author order was taken from the authors' repository citation, because the Springer author block was not readable here.
- `olmo2025olmo2`: upgraded to the shorter COLM 2025 version ("2 OLMo 2 Furious (COLM's Version)", first author Evan Pete Walsh). Its abstract covers the 7B/13B/32B family. For details that exist only in the longer `arXiv v3` report (8 October 2025), cite `arXiv:2501.00656`, which is named in the entry's note.
- `pomerleau1988alvin`: listed by NeurIPS under NIPS 1988; the Morgan Kaufmann volume appeared in 1989 (CMU Robotics Institute record, which also gives pp. 305-313). The year is kept as 1988 to match the NeurIPS listing, with a note.
- `google2026gemma4`: the official announcement is cited. A Gemma 4 Technical Report now exists (`arXiv:2607.02770`). Consider adding it if the paper cites Gemma 4 architecture details.
- `qwen2025qwen3github`, `qwen2026github`, `meta2025llamamodels`, `hf2025smollm3`, `google2025gemma3n`, `google2026gemma4`, `mistral2025*`: first-party web documentation, kept under the "official model documentation" rule. These pages can change, so access dates are recorded.
- `efron1993bootstrap`: the original imprint is New York: Chapman & Hall, 1993 (WorldCat/Open Library, LCCN 93004489). The Taylor & Francis DOI belongs to the publisher's current listing, which dates the book 1994.

#### Consequences for the paper and the review

- **In-text labels produced by `agsm`.** When two entries share an "X et al." label and year but have different author lists, `agsm.bst` disables "et al." for those entries and prints every surname (its "confusion" pass). It adds a/b suffixes only when the full surname lists are identical. With the current bib this affects `wei2022cot / wei2022emergent`, `zhou2023lima / zhou2023ifeval`, `lin2024urial / lin2024awq`, `wang2024myanswerc / wang2024lookattext`, and `liu2025mindstep / liu2025rational / liu2025quanthurts`, which will cite as, e.g., "Wei, Wang, Schuurmans, Bosma, Ichter, Xia, Chi, Le and Zhou (2022)". The three Mistral AI 2025 pages become 2025a/b/c (`mistral2025mistral3`, `mistral2025small3`, `mistral2025small31`). This is `agsm`'s behaviour, not a bib error; a different `.bst` would be needed for "et al." with year suffixes.
- **Citations whose rendered form changed.** `alain2016probes` now renders as 2017; `zheng2025qwen3quant` as Zheng et al. (2026); `deepseekai2025r1` as Guo et al. (2025); `olmo2025olmo2` as Walsh et al. (2025); `qwen2024qwen25` as Yang et al. (2024); `hf2025smollm3` as Bakouch et al. (2025); `robinson2023mcsb` as Robinson and Wingate (2023). `\citet/\citep` update automatically, but check any hand-written author names next to these keys. At audit time `paper/sections/setup.tex` cites `qwen2024qwen25`, `olmo2025olmo2` and `deepseekai2025r1`, and `benchmark.tex`, `introduction.tex` and `related_work.tex` cite `robinson2023mcsb`.
- **`docs/literature_review.md` (mechanical edit only).** A script replaced the 134 bracketed key groups in the prose with parenthetical Harvard citations using exactly the `agsm` labels above, keeping the order within each group and merging same-author runs as `natbib` does. It then appended a "Reference list" of the 106 cited works, generated from the audited bib and sorted in `agsm` order. A check confirmed that the text outside the 134 citation groups is byte-identical to the original, and re-running the script is a no-op. The ``[key]`` labels after each annotation header are inline code (identifiers linking each annotation to `refs.bib`), so they were left as they are.
- **Review prose now out of date (not edited, per instructions).** The opening note still says "106 verified BibTeX entries" (now 113). In Section 4, the "Venues not confirmed, so cited as `arXiv`" list is out of date: "Look at the Text" (COLM 2024), PAPRIKA (ICML 2025), "Quantization Hurts Reasoning?" (COLM 2025), Alain and Bengio (ICLR 2017 Workshop Track) and GameBench (NeurIPS 2024 workshop) now cite published versions, and SmoLLM2 stays `arXiv`. The "Author lists only partly confirmed" list is resolved (see table). The Reflexion line no longer matches the bib, which follows the NeurIPS author list. The Gemma 4 technical report now exists (`arXiv:2607.02770`), and its Apache-2.0 licence is confirmed by Google's announcement. OLMo 2 now cites the COLM 2025 version, and the Qwen3 quantisation study cites its 2026 Visual Intelligence version.

#### Added after the audit: decision models

Nine entries on "System One" decision models (TypeSafe's Jev and its open-weight relatives) were added on 8 October 2026 for the comparison section. They are official announcements, documentation and model cards of the releasing organisations (TypeSafe AI, Cloudflare, Perplexity, the Kev repository) plus TechCrunch and The New Stack reporting. Kev's README was read directly; the TypeSafe, Cloudflare, Perplexity and TechCrunch pages were confirmed through search-engine renderings of the pages themselves because direct fetches were blocked from the writing environment. TechCrunch bylines are not recorded and should be added when the pages are reachable: Harvard style prefers the journalist's name over the publication's.

key	source type	verified via
<code>typesafe2026jev</code>	official announcement	<code>typesafe.ai</code> blog (search rendering)
<code>typesafe2026choice</code>	official documentation	<code>docs.typesafe.ai</code> (search rendering)
<code>typesafe2026systemone</code>	official documentation	<code>docs.typesafe.ai</code> (search rendering)

key	source type	verified via
techcrunch2026jev	reporting	techcrunch.com (search rendering, dated 18 Sep 2026)
techcrunch2026decisionmodels	reporting	techcrunch.com (search rendering, dated 1 Oct 2026)
palmer2026kev	software + model cards	github.com/jaredpalmer/kev README (fetched)
cloudflare2026clef	official announcement	blog.cloudflare.com and developers.cloudflare.com changelog (search rendering)
perplexity2026decider	model card + docs	huggingface.co/perplexity-ai (search rendering)
thenewstack2026jev	reporting	thenewstack.io (search rendering)
convai2026laya	model cards + official site	huggingface.co/convaiinnovations and laya.convaiinnovations.com (search rendering)
lee2026semif	software repository	github.com/TheoLeeCJ/SemIf-OpenJev (search rendering of the README)

Two arXiv preprints that evaluate decision models (arXiv:2609.24574 on text annotation; arXiv:2609.28940 on penetration-testing harnesses, single independent author) were found but not cited: the first's authorship could not be confirmed and the second is not from an established lab, so neither meets the source rule.